

Index

Unit – 1 $\rightarrow$ Descriptive Statistics	3
1) Method 1 $\rightarrow$ Frequency Distributions	6
2) Method 2 $\rightarrow$ Measures of Central Tendency	10
Method 2.1 $\rightarrow$ Mean	12
Method 2.2 $\rightarrow$ Median	22
Method 2.3 $\rightarrow$ Mode	28
3) Method 3 $\rightarrow$ Measures of Dispersion	36
Method 3.1 $\rightarrow$ Range	39
Method 3.2 $\rightarrow$ Mean Deviation	43
Method 3.3 $\rightarrow$ Standard Deviation	48
Method 3.4 $\rightarrow$ Variance	53
Method 3.5 $\rightarrow$ Coefficient of Variation	56
4) Method 4 $\rightarrow$ Moments	63
5) Method 5 $\rightarrow$ Measures of Skewness	69
6) Method 6 $\rightarrow$ Kurtosis	76

Unit – 1 $\Rightarrow$ Descriptive Statistics

Introduction to Statistics

→ Imagine this...

- You open YouTube... and it recommends exactly what you wanted to watch
- You open Netflix... and it already knows your mood
- You type on Google... and it completes your sentence.

→ How do these systems become so smart?

- The answer is: **Statistics**.

→ In the world of Computer Science and AI-ML, we don't just "write code." We build systems that learn, predict, and decide.

- Machine Learning = The act of learning from data.
- Statistics = The logic used to understand that data.

→ Without statistics, AI is just code without intelligence. It is a brain without logic, data without meaning, and a program without a purpose.

If AI is the body, Statistics is the mind.

→ Once you start seeing the world statistically, everything—from your daily decisions to complex research—begins to make deeper, more logical sense.

→ Imagine this...You're standing in a busy city—cars rushing, people moving, signals changing, thousands of events happening every second.

→ Now someone asks: "How busy is this city, really?"

→ You can't count everything—every car, every person, every movement.

→ But you can: Collect data, find patterns and make meaningful conclusions.

→ Now think like an engineer:

- YouTube doesn't track every second of your life.
- Netflix doesn't know everything about you.

→ But they analyze patterns from data to make smart decisions.

→ That's exactly what Statistics does.

Statistics is the science of collecting, organizing, analyzing, and interpreting data to make decisions.

What is Data?

→ **Data = Any information collected for analysis / Raw facts or observations.**

→ Examples

- Marks of students: 23, 45, 67, 89, 45, 23
- Heights of players: 160 cm, 170 cm, 175 cm
- Number of likes on Instagram posts → 1.4 M. 2.5 K, 40 M
- Colors of cars: Red, Blue, White, Red

→ Data can be numbers, text, images, audio, sensor output... anything that carries information.

How Data Convert into Frequency?

→ **Frequency = How many times a value appears in the data.**

→ Data alone looks messy.

- Frequency helps us *count* how often each value occurs.

→ Example

- Suppose data = 23, 45, 23, 67, 45, 23, 89

Value	Frequency
23	3 Times
45	2 Times
67	1 Time
89	1 Time

- So here, **frequency** is just the count of repetition.

Origin of the word “Statistics”

→ The word comes from:

- **Latin:** status → meaning *state* or *condition*
- **Italian:** statista → meaning *statesman* – a person who runs the government
- **German:** statistik → meaning *science of the state*

→ It referred to the **systematic collection of facts** for the government:

Unit – 1 Descriptive Statistics

- Births
 - Deaths
 - Land records
- Originally, statistics was used by kings and governments to understand population, wealth, land, and taxes.
- There are two types of statistics:
- (1) Descriptive Statistics (Unit – 1)
 - Descriptive statistics refers to the quantitative methods used to **summarize** a set of observations.
 - This includes mean, median, mode, range, variance, standard deviation, etc.
 - (2) Inferential Statistic (Unit – 4 & 5)
 - Inferential statistics **draw conclusions** and **make predictions** about a population based on a sample using techniques like hypothesis testing and regression.

Descriptive Statistics

- **Descriptive Statistics** is a branch of statistics that **describes and summarizes data** in a simple and meaningful way.
- It does NOT make predictions or generalizations.
 - It simply tells what the data is showing.
- It answers questions like:
- What is the average score of the class?
 - What is the highest and lowest mark?
 - How spread out are the marks?

Method 1 $\rightsquigarrow$ Frequency Distributions

Types of Data

→ Ungrouped / Row Data

- Data in its original form.
- Observations are listed individually.
- Example: 10, 25, 26, 35, 3, 8, 19, 29, 30, 18

→ Discrete Data

- Data that can take only distinct, countable values. Usually obtained by counting.
- Examples
 - Number of students in a class
 - Number of books in a library
 - Number of vehicles passing a toll booth
- **Representation:**

x_i	1	2	3	4	5	6
f_i	4	7	8	9	10	2

→ Continuous Data

- Data that can take any value within a range.
- Usually obtained by measurement.
- Examples: Height, Weight, Temperature, Time
- **Representation:**

Class Interval	0 – 2	3 – 5	6 – 8	9 – 11
Frequency	11	7	8	0

Unit – 1 Descriptive Statistics

Frequency Distribution

→ A **frequency distribution** is a method of organizing data that shows how often each value or group of values occurs in a dataset.

→ Types of Frequency Distribution:

- **Ungrouped** Frequency Distribution (**Discrete** Frequency Distribution)
- **Grouped** Frequency Distribution (**Continuous** Frequency Distribution)

Frequency Distribution of Ungrouped Data

→ **Discrete Frequency Distribution**

- A discrete frequency distribution shows each value and how many times it appears in the data.
- For Example: Data of students using library during exam time.

Reading Hours (x_i)	1	2	3	4	5	6
No. of students (f_i)	4	7	8	9	10	2

Frequency Distribution of Grouped Data

→ **Continuous Frequency Distribution**

- A continuous frequency distribution groups data into **class intervals** and shows how many values fall in each interval.
- When the number of observations is large, individual values are grouped into class intervals to simplify analysis and interpretation.
- For Example: Reading hours of students during exam time

Reading Hours (x_i)	0 – 2	3 – 5	6 – 8	9 – 11
No. of students (f_i)	11	27	2	0

Lower Bound & Upper Bound

→ For the class $x_i - x_{i+1}$,

- lower bound is x_i and
- upper bound is x_{i+1} .
- In above example, for the class interval 0 – 2

- Lower bound = 0 and Upper bound = 2

→ There are two types of class intervals:

Exclusive Class Interval

- In an exclusive class interval, the upper bound of one class is not included in that class.
- Example: **0 – 2, 2 – 4, 4 – 6, ...**
 - Here, 2 belongs to the next class **2 – 4**, not to **0 – 2**.

Inclusive Class Interval

- In an inclusive class interval, both lower and upper bounds are included in the same class.
- Example: **0 – 2, 3 – 5, 6 – 8, ...**
- Inclusive classes are converted into exclusive classes to remove gaps between class intervals and make the data continuous.

→ **To convert inclusive classes into exclusive classes, follow these steps:**

- **Step 1:** Find Value of Gap

$$\text{Gap} = \frac{\text{Lower bound of next class} - \text{Upper bound of current class}}{2}$$

- **Step 2:** Adjust Class Intervals
 - Subtract gap from the lower bound and add gap to the upper bound.
- For Example:

Inclusive Class	0 – 2	3 – 5	6 – 8
Exclusive Class	–	–	–

Step 1: Find Value of Gap

$$\text{Gap} = \frac{3 - 2}{2} = 0.5$$

Step 2: Adjust Class Intervals

Subtract gap from the lower bound: $0 - 0.5 = -0.5$

add gap to the upper bound: $2 + 0.5 = 2.5$

Unit – 1 Descriptive Statistics

Inclusive Class	0 – 2	3 – 5	6 – 8
Exclusive Class	–0.5 – 2.5	2.5 – 5.5	5.5 – 8.5

Class Length

→ A difference between Upper boundary and Lower boundary is known as class length.

→ It is denoted by **c** and defined as

$$c = \text{Upper Boundary} - \text{Lower Boundary} = x_{i+1} - x_i$$

→ For Example:

- For the class 3 – 5, class length is $5 - 3 = 2$.

Mid-Point of Class

→ Mid-point of class is used to find x_i in case of continuous frequency distribution.

→ It is defined as follow:

$$\text{Mid point of class} = x_i = \frac{\text{Lower Boundary} + \text{Upper Boundary}}{2}$$

→ For Example:

$$\text{For the class 3 – 5, mid point is } \frac{3 + 5}{2} = 4.$$

Method 2 $\rightsquigarrow$ Measures of Central Tendency

Introduction

→ Imagine you have a group of numbers.

- Every day, technology companies collect millions of data values:
 - Netflix records watch times
 - Instagram tracks likes
 - Google Maps measures traffic speed
 - Swiggy records delivery times
 - AI models generate prediction scores

→ Now imagine reading every single number one by one is Impossible.

→ So, we ask a simple question:

Can one value summarize the overall behavior of the data?

→ That representative value is called the **Measure of Central Tendency**.

- It tells us where the center of the data lies.

→ Centre → the middle or balancing point

→ Tendency → where the data values tend to gather

Why do we need Measure of Central Tendency?

→ Real-world data is often huge, scattered, and difficult to understand.

→ Instead of analyzing every value individually, we use one representative value to summarize the data.

→ **Example:**

- Netflix records daily watch times (in minutes) of users: 90, 100, 95, 105, 98.
- Instead of studying each value separately, Netflix may simply say: Users generally watch around 100 minutes per day.
- Here, 100 minutes represents the typical behavior of users because most values are close to it.

→ Thus, a measure of central tendency helps us summarize large data into one meaningful value.

Unit – 1 Descriptive Statistics

What is a Measure of Central Tendency?

- A Measure of Central Tendency is a value (or values) that represents the center or typical behavior of a dataset.
- Here, “typical behavior” means the value that best describes how the data usually behaves or where most data values tend to concentrate.
- There are three main measures to estimate central tendency which is
- (1) Mean ($\bar{x}$)
 - (2) Median (M)
 - (3) Mode (Z)

Method 2.1 $\rightsquigarrow$ Mean

→ The mean or the arithmetic average is the single value that every observation would have if the total were distributed equally among all observations.

→ Example

- Data: 12, 13, 15, 16 and Total =56
- If we replace all observations by one common value without changing the total:

14, 14, 14, 14

- Total is still 56. So, Mean =14

→ Here, 14 is not just a calculation; it is the fair-share value of a dataset.

- It is denoted by “ $\bar{x}$ ” and read as “x bar”.

→ Mean is a **measure of central tendency** because it provides a single representative value for the whole dataset.

→ Table of different formulae of mean is as follow:

Method	Ungrouped Data	Discrete Frequency Data	Continuous Frequency Data
Direct Method	$\frac{\sum x_i}{n}$	$\frac{\sum f_i x_i}{\sum f_i}$	

- x_i = the individual observations
 - x_i = **midpoints of class** intervals in Continuous Frequency Data
- n = **total number of observations** in ungrouped data.
- f_i = the **frequency** of x_i
- $\sum x_i$ = sum of all observations
- $\sum f_i x_i$ = sum of product of observations and frequencies
- $\sum f_i$ = sum of frequencies

Unit – 1 Descriptive Statistics

Method	Ungrouped Data	Discrete Frequency Data	Continuous Frequency Data
Assumed Mean Method	$A + \frac{\sum d_i}{n}$	$A + \frac{\sum f_i d_i}{\sum f_i}$	

- Here, A can be any value from x_i .
 - It is known as assumed mean value
- $d_i = x_i - A$
 - d_i is called the **deviation** of an observation from the assumed mean
- $\sum d_i$ = sum of all deviations
- $\sum f_i d_i$ = sum of product of deviations and frequencies
- $\sum f_i$ = sum of frequencies

Method	Ungrouped Data	Discrete Frequency Data	Continuous Frequency Data
Step Deviation Method	-----	-----	$A + \frac{\sum f_i u_i}{\sum f_i} \times c$

- c = Class Length

$$u_i = \frac{x_i - A}{c}$$

- u_i is called the **step deviation** of an observation from the assumed mean.
- $\sum f_i u_i$ = sum of product of step deviations and frequencies
- $\sum f_i$ = sum of frequencies

Examples of Method - 2.1: Mean

C	1	Find the mean of data 10.2, 9.5, 8.3, 9.7, 9.5, 11.1, 7.8, 8.8, 9.5, 10. Answer: 9.44																		
H	2	Find mean of following data: (a) 2, 8, 4, 6, 10, 12, 4, 8, 14, 16 (b) 10, 9, 21, 16, 14, 18, 20, 18, 14, 18, 23, 16, 18, 4 Answer: (a) 8.4, (b) 15.6429																		
T	3	A first-year CSE student solved coding problems on an online platform over 5 days: 3, 5, 4, 6, 7 Find the average number of problems solved per day. Also determine, on which days did the student perform above average? Answer: Average problems solved per day = 5, The student performed above average on Day 4 and Day 5.																		
C	4	Find the mean for following data: <table border="1"><tr><td>Marks obtained</td><td>20</td><td>9</td><td>25</td><td>50</td><td>40</td><td>80</td></tr><tr><td>Number of students</td><td>6</td><td>4</td><td>16</td><td>7</td><td>8</td><td>2</td></tr></table> Answer: 32.23	Marks obtained	20	9	25	50	40	80	Number of students	6	4	16	7	8	2				
Marks obtained	20	9	25	50	40	80														
Number of students	6	4	16	7	8	2														
H	5	Find the mean for following data: <table border="1"><tr><td>Weight of students</td><td>18</td><td>22</td><td>30</td><td>35</td><td>39</td><td>42</td><td>45</td><td>47</td></tr><tr><td>Number of students</td><td>4</td><td>5</td><td>8</td><td>8</td><td>16</td><td>4</td><td>2</td><td>3</td></tr></table> Answer: 34.5	Weight of students	18	22	30	35	39	42	45	47	Number of students	4	5	8	8	16	4	2	3
Weight of students	18	22	30	35	39	42	45	47												
Number of students	4	5	8	8	16	4	2	3												

C	6	Find the mean using direct method, assumed mean method and step deviation method: <table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td></tr><tr><td>No. of students</td><td>5</td><td>10</td><td>40</td></tr><tr><td></td><td>30 – 40</td><td>40 – 50</td><td></td></tr><tr><td></td><td>20</td><td>25</td><td></td></tr></table> Answer: 30	Marks	0 – 10	10 – 20	20 – 30	No. of students	5	10	40		30 – 40	40 – 50			20	25	
Marks	0 – 10	10 – 20	20 – 30															
No. of students	5	10	40															
	30 – 40	40 – 50																
	20	25																
H	7	The following data represents the no. of foreign visitors in a multinational company in every 10 days during last 2 months. Use the data to find the mean. <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td></tr><tr><td>No. of visitors</td><td>12</td><td>18</td><td>27</td><td>20</td></tr></table> Answer: 28	Class	0 – 10	10 – 20	20 – 30	30 – 40	No. of visitors	12	18	27	20						
Class	0 – 10	10 – 20	20 – 30	30 – 40														
No. of visitors	12	18	27	20														
T	8	The training time of ML models (in minutes) on different systems is given below: <table><tr><td>Training Time (min)</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>Number of Models</td><td>6</td><td>10</td><td>14</td><td>8</td><td>2</td></tr></table> Find (1) The arithmetic mean training time. (2) Which class interval contributes most to the mean? (3) How can average training time help improve AI infrastructure? Answer: (1) 32.5 Minutes, (2) 30 – 40 minutes (3) Average training time helps companies use computers and servers more efficiently. It also helps save time and reduce cost while training ML models.	Training Time (min)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	Number of Models	6	10	14	8	2				
Training Time (min)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60													
Number of Models	6	10	14	8	2													

T	9	A streaming platform analyzed the daily watch time of users watching a trending web series. <table><tr><td>Daily Watch Time (minutes)</td><td>0 – 9</td><td>10 – 19</td><td>20 – 29</td><td>30 – 39</td></tr><tr><td>Number of Users</td><td>5</td><td>8</td><td>12</td><td>10</td></tr></table> Find the arithmetic mean watch time using: (1) Direct Method (2) Assumed Mean Method (3) Step Deviation Method Answer: 22.21 Minutes	Daily Watch Time (minutes)	0 – 9	10 – 19	20 – 29	30 – 39	Number of Users	5	8	12	10				
Daily Watch Time (minutes)	0 – 9	10 – 19	20 – 29	30 – 39												
Number of Users	5	8	12	10												
C	10	Find the missing frequency f_1 and f_2 in the table given below, it is being given that the mean of the given frequency distribution is 50. <table><tr><td>Class</td><td>0 – 20</td><td>20 – 40</td><td>40 – 60</td><td>60 – 80</td><td>80 – 100</td><td>Total</td></tr><tr><td>f</td><td>17</td><td>f_1</td><td>32</td><td>f_2</td><td>19</td><td>100</td></tr></table> Answer: $f_1 = 18$, $f_2 = 14$	Class	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	Total	f	17	f_1	32	f_2	19	100
Class	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	Total										
f	17	f_1	32	f_2	19	100										
T	11	An esports platform recorded the number of kills made by players in matches. <table><tr><td>Kills (x)</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Number of Players (f)</td><td>3</td><td>7</td><td>x</td><td>6</td><td>2</td></tr></table> The arithmetic mean kills per player was 5.8. Find the missing frequency x. Also answer: Which kill score contributes most to the average? Note: Contributes most to the average means which value has the greatest effect on the mean. Answer: $x = 12$, Kill score 6 contributes most to the average because its fx value is maximum.	Kills (x)	2	4	6	8	10	Number of Players (f)	3	7	x	6	2		
Kills (x)	2	4	6	8	10											
Number of Players (f)	3	7	x	6	2											

Unit – 1 Descriptive Statistics

H	12	Find the missing frequency from the following data if mean is 19.92. <table><tr><td>Class</td><td>4 – 8</td><td>8 – 12</td><td>12 – 16</td><td>16 – 20</td><td>20 – 24</td></tr><tr><td>f</td><td>11</td><td>13</td><td>16</td><td>14</td><td>?</td></tr><tr><td></td><td>24 – 28</td><td>28 – 32</td><td>32 – 36</td><td>36 – 40</td><td></td></tr><tr><td></td><td>9</td><td>17</td><td>6</td><td>4</td><td></td></tr></table> Answer: 10	Class	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24	f	11	13	16	14	?		24 – 28	28 – 32	32 – 36	36 – 40			9	17	6	4	
Class	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24																					
f	11	13	16	14	?																					
	24 – 28	28 – 32	32 – 36	36 – 40																						
	9	17	6	4																						
T	13	Find the missing frequency from the following data if mean is 30. <table><tr><td>x</td><td>10-14</td><td>15-19</td><td>20-24</td><td>25-29</td><td>30-34</td><td>35-39</td><td>40-44</td><td>45-49</td></tr><tr><td>f</td><td>4</td><td>6</td><td>10</td><td>5</td><td>f</td><td>3</td><td>9</td><td>6</td></tr></table> Answer: f = 7	x	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49	f	4	6	10	5	f	3	9	6						
x	10-14	15-19	20-24	25-29	30-34	35-39	40-44	45-49																		
f	4	6	10	5	f	3	9	6																		
T	14	A music streaming app analyzed the daily listening time of users for a popular Spotify playlist. <table><tr><td>Daily Watch Time (minutes)</td><td>0 – 9</td><td>10 – 19</td><td>20 – 29</td><td>30 – 39</td></tr><tr><td>Number of Users</td><td>5</td><td>8</td><td>x</td><td>y</td></tr></table> The total number of users is 50 and the arithmetic mean listening time is 24.5 minutes. Find the missing frequencies x and y. Also answer: (1) Which listening-time interval is closest to the average? (2) Why is average listening time important for Spotify’s recommendation system? (3) How can missing data affect music recommendation algorithms? Answer: x = 19, y = 18, (1) 20 – 29 minutes	Daily Watch Time (minutes)	0 – 9	10 – 19	20 – 29	30 – 39	Number of Users	5	8	x	y														
Daily Watch Time (minutes)	0 – 9	10 – 19	20 – 29	30 – 39																						
Number of Users	5	8	x	y																						
C	15	A co-operative bank has two branches employing 50 and 70 workers respectively. The average salaries paid by two respective branches are 360 and 390 rupees per month. Calculate the mean of the salaries of all the employees. Answer: 377.5																								

H	16	In a factory, there are 70 workers, out of whom 20 are females. The average daily wage of female worker is 325 Rs. The total daily wages of all the workers are 25,900 Rs. Determine the daily wages of a male worker. Answer: 388 Rs.									
H	17	Two influencers uploaded reels on Instagram. <table border="1"> <thead> <tr> <th>Influencer</th><th>Number of Reels</th><th>Average Likes per Reel</th></tr> </thead> <tbody> <tr> <td>Creator A</td><td>25</td><td>18,000</td></tr> <tr> <td>Creator B</td><td>40</td><td>24,000</td></tr> </tbody> </table> Calculate the combined average likes per reel for both creators. Answer: 21,692.31 likes	Influencer	Number of Reels	Average Likes per Reel	Creator A	25	18,000	Creator B	40	24,000
Influencer	Number of Reels	Average Likes per Reel									
Creator A	25	18,000									
Creator B	40	24,000									
T	18	The mean of 20 observations is 15. One observation 25 was wrongly taken as 35 while calculating the mean. What is the correct mean? Answer: 14.5									
T	19	The mean of 50 observations was calculated as 62. Later, it was discovered that two observations were wrongly recorded: - One value 48 was recorded as 84 - Another value 96 was recorded as 69 Find the correct mean. Answer: 61.82									
T	20	A streaming platform has two types of subscribers. 800 Basic users watch content for an average of 18 hours per month. 500 Premium users watch content for an average of 32 hours per month. During a festival season, 200 new Premium users joined and their average watch time was 40 hours per month. Find the new combined average watch time of all subscribers. Answer: 25.6 hours per month									

T	21	The mean of 9 numbers is 40. If one number is excluded, the mean becomes 38. Find the excluded number. Answer: 56
T	22	The mean of 15 observations is 32. After excluding two observations, the mean of the remaining 13 observations becomes 30. If one excluded observation is 50, find the other. Answer: 48 and 60
H	23	Give one advantage and one limitation of arithmetic mean.
T	24	Why is arithmetic mean called a measure of central tendency?
H	25	(1) If each observation in a data set is increased by 5, what happens to the arithmetic mean? (2) If every value in a data set is multiplied by 3, how does the mean change? Answer: (1) The mean increases by 5, (2) The mean gets multiplied by 3
T	26	Two content creators uploaded videos. • Creator A uploaded 4 videos with average views of 80,000. • Creator B uploaded 5 videos with average views of 70,000. Suddenly, one new viral video with 300,000 views is added to Creator A. (1) Find Creator A's new average views. (2) Compare the new average with Creator B. (3) By how much did Creator A's average increase? Answer: (1) 1, 24, 000 Views, (2) Creator A's new average is 54, 000 views higher than Creator B's average. (3) 44, 000 views.

T**27**

A gaming platform recorded the number of live viewers joining esports streams per minute.

Viewers per Minute (x)	100	200	300	400	500
Frequency(f)	5	8	15	10	6

After calculating the arithmetic mean, engineers discovered:

- Frequency for 500 viewers should be 2 instead of 6
- Frequency for 100 viewers should be 7 instead of 5

Find:

- (1) The incorrect mean viewers per minute.
- (2) The corrected mean.
- (3) How much did the average change?

Answer: (1) 309.09 viewers, (2) 280.95 viewers,

(3) the average decreased by 28.14 viewers

T	28	A music streaming company analyzed daily listening time of users.					
		Listening Time (min)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
		Number of Users	8	15	20	12	10

Later, the analytics team found:

Frequency of class 80–100 should be 4 instead of 10,
 Frequency of class 60–80 should be 9 instead of 12,
 Frequency of class 0–20 should be 10 instead of 8.

Find:

- (1) The incorrect arithmetic mean listening time.
- (2) The corrected arithmetic mean.
- (3) Which interval contributed most to the error?
- (4) Explain how wrong listening-time data can affect AI music recommendations.

Answer: (1) Incorrected Mean = 50.31 Minutes,

(2) Correted Mean = 43.79 Minutes,

(3) The interval 80 – 100 min. contributed most to the error

(4) Wrong listening – time data can make the AI system misunderstand user music preferences. As a result, users may get less suitable song recommendations.

Method 2.2 $\rightsquigarrow$ Median

- The median is the value found at the **exact middle** of the set of values.
- Median is denoted by capital letter “**M**”.
- To compute the median, list all observations in ascending (or descending) order and then locate the value in the center of the sample.
- Cumulative Frequency:
 - Cumulative frequency cf_i is the sum of all previous frequencies up to a given observation or class.
 - For Example:

x_i	f_i	cf_i
9	5	5
20	4	$5 + 4 = 9$
25	3	$5 + 4 + 3 = 12$

9 9 9 9 9

First 5 observations are 9

20 20 20 20

Observations 6th to 9th are 20

Observations 10th to 12th are 25

- Table of formula of median.

Data	Formula
Ungrouped Data	If n is odd , then $M = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ observation}$
Discrete Frequency Data	If n is even , then $M = \frac{\left(\frac{n}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ observation}}{2}$
Continuous Frequency Data	$M = L + \left(\frac{\frac{n}{2} - F}{f} \right) \times c$

Unit – 1 Descriptive Statistics

→ Where,

Median class = Class whose cumulative frequency with property $\min \left\{ cf \mid cf \geq \frac{n}{2} \right\}$

L = Lower boundary point of the median class

n = Total number of observation (sum of the frequencies)

F = Cumulative frequency of the class preceding the median class

f = The frequency of the median class

Example of Method - 2.2: Median

C	1	Find the median of following data: 20, 25, 30, 15, 17, 35, 26, 18, 40, 45, 50. Answer: 26
H	2	Find the median of following data: (a) 6, 20, 43, 50, 19, 53, 0, 37, 78, 1, 15. (b) 10, 34, 27, 24, 12, 27, 20, 18, 15, 30. (c) 110, 115, 108, 112, 120, 116, 140, 135, 128, 132. Answer: (a) 20, (b) 22, (c) 118
T	3	Views (in thousands) of viral shorts: 18, 22, 24, 25, 27, 30, 120 Find the median views and comment why median is better than mean here. Answer: Median Views = 25 Thousand
C	4	The given observations have been arranged in ascending order. If the median of the data is 63, find the value of x for the following data: 29, 32, 48, 50, x, x + 2, 72, 78, 84, 95. Answer: x = 62
H	5	If the median of the data is 2, find the value of a: -9, -4, a, 5, 8, 11. Answer: - 1

H	6	The annual salary packages (in lakh ₹) of 10 students placed in a software company are arranged in ascending order: $3.5, 4.2, 5.0, 5.8, x, x + 1.5, 12, 15, 18, 25$ If the median package is 8.4 lakh ₹, find the value of x. Answer: $x = 7.65$ lakh ₹																
T	7	A cloud company monitored the server recovery times (in minutes) after cyberattacks. The ordered observations are: $12, 18, 25, x - 4, x + 1, x + 7, x + 12, 96, 120, 150$ If the median recovery time is 58 minutes, find all possible values of x. Answer: $x = 54$																
T	8	A sorted dataset contains 10 observations: $x - 12, x - 5, x - 2, x, x + 3, x + 7, x + 10, x + 14, x + 20, x + 30$ If the median is 52, find the value of x. Then find: 1) Mean 2) Whether mean > median or not Answer: $x = 47$, (1) $\bar{x} = 53.5$, (2) Mean > Median																
C	9	Calculate the median for the following data: <table border="1"><tr><td>Marks</td><td>20</td><td>9</td><td>25</td><td>50</td><td>40</td><td>80</td></tr><tr><td>No. of students</td><td>6</td><td>4</td><td>16</td><td>7</td><td>8</td><td>2</td></tr></table> Answer: 25	Marks	20	9	25	50	40	80	No. of students	6	4	16	7	8	2		
Marks	20	9	25	50	40	80												
No. of students	6	4	16	7	8	2												
H	10	Obtain the median size of shoes sold from the following data: <table border="1"><tr><td>Size</td><td>5</td><td>5.5</td><td>6</td><td>6.5</td><td>7</td><td>7.5</td><td>8</td></tr><tr><td>Pair</td><td>30</td><td>40</td><td>50</td><td>150</td><td>300</td><td>600</td><td>950</td></tr></table> Answer: 7.5	Size	5	5.5	6	6.5	7	7.5	8	Pair	30	40	50	150	300	600	950
Size	5	5.5	6	6.5	7	7.5	8											
Pair	30	40	50	150	300	600	950											

T	11	A music streaming company recorded the number of hours users spent listening to music during a festival week. <table><tr><td>Listening Hours</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Number of Users</td><td>$x - 3$</td><td>$x + 1$</td><td>$2x$</td><td>$3x - 2$</td><td>$x + 4$</td></tr></table> If the median listening time is 4 hours, find all possible values of x. Further: 1) Find the total number of users (N) 2) Check whether the median remains 4 if 20 more users are added to the “5 hours” category Answer: $x = 3$, $N = 24$, the median increases to 5 hours.	Listening Hours	1	2	3	4	5	Number of Users	$x - 3$	$x + 1$	$2x$	$3x - 2$	$x + 4$
Listening Hours	1	2	3	4	5									
Number of Users	$x - 3$	$x + 1$	$2x$	$3x - 2$	$x + 4$									
C	12	The following table gives marks obtained by 50 students in statistics. Find the median. <table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>No. of students</td><td>16</td><td>12</td><td>18</td><td>3</td><td>1</td></tr></table> Answer: 17.5	Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	No. of students	16	12	18	3	1
Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50									
No. of students	16	12	18	3	1									
H	13	A startup company tracked the daily screen time (in hours) of employees using productivity-monitoring software during a work-from-home week. The collected data is given below: <table><tr><td>Screen Time (hrs.)</td><td>0 – 2</td><td>2 – 4</td><td>4 – 6</td><td>6 – 8</td><td>8 – 10</td></tr><tr><td>Number of Employees</td><td>28</td><td>12</td><td>7</td><td>2</td><td>1</td></tr></table> Find the median screen time of employees. Answer: Median screen time ≈ 1.79 hours Interpretation: This means that about 50% of employees spent less than 1.79 hours on screen time, while the other 50% spent more.	Screen Time (hrs.)	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10	Number of Employees	28	12	7	2	1
Screen Time (hrs.)	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10									
Number of Employees	28	12	7	2	1									

T	14	The following table gives the marks obtained by 50 students in mathematics. Find the median. <table><tr><td>x</td><td>10-14</td><td>15-19</td><td>20-24</td><td>25-29</td><td>30-34</td><td>35-39</td></tr><tr><td>f</td><td>4</td><td>6</td><td>10</td><td>5</td><td>7</td><td>3</td></tr><tr><td></td><td>40-44</td><td>45-49</td><td colspan="4"></td></tr><tr><td></td><td>9</td><td>6</td><td colspan="4"></td></tr></table> Answer: 29.5	x	10-14	15-19	20-24	25-29	30-34	35-39	f	4	6	10	5	7	3		40-44	45-49						9	6				
x	10-14	15-19	20-24	25-29	30-34	35-39																								
f	4	6	10	5	7	3																								
	40-44	45-49																												
	9	6																												
C	15	The median of 60 observations (following data) is 28.5. Find x and y. <table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>No. of students</td><td>5</td><td>x</td><td>20</td><td>15</td><td>y</td><td>5</td></tr></table> Answer: x = 8, y = 7	Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	No. of students	5	x	20	15	y	5														
Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60																								
No. of students	5	x	20	15	y	5																								
H	16	Calculate the missing frequency from the following distribution, it is being given that the median of the distribution is 24. <table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>No. of students</td><td>5</td><td>25</td><td>x</td><td>18</td><td>7</td></tr></table> Answer: x = 25	Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	No. of students	5	25	x	18	7																
Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50																									
No. of students	5	25	x	18	7																									
T	17	Netflix recorded the weekly watching hours of users during a holiday season: <table><tr><td>Watching Hours</td><td>0 – 5</td><td>5 – 10</td><td>10 – 15</td><td>15 – 20</td><td>20 – 25</td></tr><tr><td>Users</td><td>3</td><td>5</td><td>7</td><td>10</td><td>35</td></tr></table> Find the median watching hours. What does the median class indicate about binge-watching habits? Answer: Median watching time $\approx$ 20.71 hours.	Watching Hours	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25	Users	3	5	7	10	35																
Watching Hours	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25																									
Users	3	5	7	10	35																									
T	18	The median of a discrete distribution is 18. Explain whether the following statement is true or false: “Exactly half of the observations are equal to 18.” Hint: The statement is false.																												
T	19	Why is cumulative frequency necessary for finding the median in discrete data?																												

Unit – 1 Descriptive Statistics

H	20	Can the median lie outside the data range? Explain with reason.
T	21	Two datasets have the same median but very different observations. Is it possible? Explain with example.
H	22	Can the median be found without arranging the data? Explain.
T	23	Why is the median preferred over the arithmetic mean in income distribution data?
C	24	An IT company reports that the average package offered to employees is ₹10 LPA. After investigation, it is found that only a few employees receive packages above ₹40 LPA, while most employees receive packages between ₹4–6 LPA. Is the arithmetic mean misleading here? Which measure should students look at to understand the typical package offered? Answer: Yes the arithmetic mean is misleading due to a few very high packages. The median should be used to understand the typical package offered.
T	25	A real estate website says: “The average house price in the city is ₹1.5 crore.” But most houses cost around ₹40–50 lakh, while a few luxury houses cost several crores. Why might the arithmetic mean be misleading here? Which measure would better represent the typical house price?

Method 2.3 $\rightsquigarrow$ Mode

Why do we need Mode?

- Mean gives the average.
- Median gives the middle value.
- But sometimes we want to know:
 - What is the most common behavior?
 - Which item is most popular?
 - Which category occurs most often?
- That is where **Mode** becomes useful.

Mode

- The mode is the **most frequently** occurring value in a dataset.
- Mode is denoted by capital letter "**Z**".
- The mode is **not** necessarily **unique**, like mean and median.
- Data set with one mode is known as **unimodal**.
- We can have data with two modes (**bi-modal**) or more than two modes (**multi-modal**).
- If frequency of all the observations is equal then, mode does not exist.

Special Property of Mode

- Mode is the only measure of central tendency that can be used for:
 - Numerical data
 - Categorical data
 - Text labels
 - Classes/groups
- **Example**

Favorite Programming Language	Students
Python	42
Java	25
C++	18

Unit – 1 Descriptive Statistics

→ Here:

- Mean cannot be calculated
- Median also makes no sense

→ But, Mode = Python because it is the most preferred language.

→ Table of formulae of mode.

Data	Formula
Ungrouped Data	The observation that occurs most frequently in the dataset.
Discrete Frequency Data	The observation corresponding to the highest frequency.
Continuous Frequency Data	$Z = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times c$

→ where,

Modal class = A class with highest frequency

L = Lower boundary of modal class

c = Class length

f_1 = Frequency of the modal class

f_0 = Frequency of the class before the modal class

f_2 = Frequency of the class after the modal class

Empirical Relation Between Mean, Median and Mode

→ The relation between the mean, median, and mode is given by the formula:

$$Z = 3M - 2\bar{x}$$

where, $\bar{x}$ = Mean, M = Median, Z = Mode

→ It helps to **estimate** one measure of central tendency when the other two are known.

→ This relation is only **useful for unimodal** frequency distributions.

Example of Method - 2.3: Mode

C	1	Find the mode of following data: (a) 2, 4, 2, 5, 7, 2, 8, 9. (b) 2, 8, 4, 6, 10, 12, 4, 8, 14, 16. Answer: (a) 2, (b) 4 & 8										
H	2	1) What is bimodal data? 2) What is multimodal data?										
H	3	Determine whether the data is unimodal, bimodal, or multimodal: 2, 2, 3, 3, 4, 4, 5										
H	4	Find the mode: 7, 8, 9, 10, 11 Answer: Mode does not exists.										
H	5	A college records most preferred AI tool: <table border="1"><thead><tr><th>Tool</th><th>Students</th></tr></thead><tbody><tr><td>ChatGPT</td><td>180</td></tr><tr><td>Gemini</td><td>75</td></tr><tr><td>Claude</td><td>45</td></tr></tbody></table> Find the mode. Answer: Z = ChatGPT	Tool	Students	ChatGPT	180	Gemini	75	Claude	45		
Tool	Students											
ChatGPT	180											
Gemini	75											
Claude	45											
T	6	Construct a dataset of 10 observations having: Mode = 15, Median = 14										
T	7	Why is Mode considered unstable in some datasets?										
C	8	Find the mode of following data: <table border="1"><thead><tr><th>x</th><th>11</th><th>22</th><th>33</th><th>44</th></tr></thead><tbody><tr><th>f</th><td>15</td><td>20</td><td>19</td><td>10</td></tr></tbody></table> Answer: 22	x	11	22	33	44	f	15	20	19	10
x	11	22	33	44								
f	15	20	19	10								

Unit – 1 Descriptive Statistics

H	9	Find the mode from the following frequency distribution: <table><tr><td>x</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>f</td><td>5</td><td>6</td><td>8</td><td>7</td><td>9</td><td>8</td><td>9</td><td>6</td></tr></table> Answer: 12 & 14	x	8	9	10	11	12	13	14	15	f	5	6	8	7	9	8	9	6						
x	8	9	10	11	12	13	14	15																		
f	5	6	8	7	9	8	9	6																		
C	10	Find the mode of following data: <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>f</td><td>3</td><td>5</td><td>7</td><td>10</td><td>12</td></tr><tr><td></td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td><td>80 – 90</td><td>90 – 100</td></tr><tr><td></td><td>15</td><td>12</td><td>6</td><td>2</td><td>8</td></tr></table> Answer: 55	Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	f	3	5	7	10	12		50 – 60	60 – 70	70 – 80	80 – 90	90 – 100		15	12	6	2	8
Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50																					
f	3	5	7	10	12																					
	50 – 60	60 – 70	70 – 80	80 – 90	90 – 100																					
	15	12	6	2	8																					
T	11	Find the mode of following data: <table><tr><td>Class</td><td>400 – 500</td><td>500 – 600</td><td>600 – 700</td></tr><tr><td>f</td><td>8</td><td>16</td><td>20</td></tr><tr><td></td><td>700 – 800</td><td>800 – 900</td><td></td></tr><tr><td></td><td>17</td><td>3</td><td></td></tr></table> Answer: 657.14	Class	400 – 500	500 – 600	600 – 700	f	8	16	20		700 – 800	800 – 900			17	3									
Class	400 – 500	500 – 600	600 – 700																							
f	8	16	20																							
	700 – 800	800 – 900																								
	17	3																								
C	12	Obtain the mean, mode and median for the following information: <table><tr><td>x</td><td>< 10</td><td>< 20</td><td>< 30</td><td>< 40</td><td>< 50</td><td>< 60</td></tr><tr><td>f</td><td>12</td><td>30</td><td>57</td><td>77</td><td>94</td><td>100</td></tr></table> Answer: $\bar{x} = 28$, $M = 27.41$, $Z = 25.63$	x	< 10	< 20	< 30	< 40	< 50	< 60	f	12	30	57	77	94	100										
x	< 10	< 20	< 30	< 40	< 50	< 60																				
f	12	30	57	77	94	100																				
T	13	Obtain the mean, median and mode for the following information: <table><tr><td>Marks</td><td>> 0</td><td>> 10</td><td>> 20</td><td>> 30</td><td>> 40</td><td>> 50</td><td>> 60</td><td>> 70</td></tr><tr><td>No. of Std.</td><td>150</td><td>140</td><td>100</td><td>80</td><td>80</td><td>70</td><td>30</td><td>14</td></tr></table> Answer: $\bar{x} = 39.27$, $M = 45$, $Z = 16$	Marks	> 0	> 10	> 20	> 30	> 40	> 50	> 60	> 70	No. of Std.	150	140	100	80	80	70	30	14						
Marks	> 0	> 10	> 20	> 30	> 40	> 50	> 60	> 70																		
No. of Std.	150	140	100	80	80	70	30	14																		

T 14

A multinational cybersecurity company uses an AI-based monitoring system to detect unusual traffic on its cloud servers. The AI system records the number of suspicious login requests (in thousands) received during different time intervals. The following frequency distribution was observed:

API Requests (x)	Frequency (f)
5	8
10	$x + 6$
15	22
20	$2x - 4$
25	30
30	$x + 18$
35	30

If the distribution is trimodal, find:

- 1) the value of x
- 2) all modal values
- 3) the common modal frequency
- 4) the total frequency of the distribution.
- 5) Based on the above data, what can the cybersecurity team conclude about:
 - attack intensity,
 - user activity patterns,
 - and server traffic behaviour?
- 6) Why might a trimodal distribution be dangerous for a cybersecurity monitoring system?
- 7) Explain why Mode is more informative than Mean in this situation.
- 8) If the frequency corresponding to 35 thousand requests suddenly increases far beyond all other frequencies, what possible cyber threat might the AI system suspect?

Unit – 1 Descriptive Statistics

H	15	The mode of the following data is 67. Find the missing frequency x.																			
		<table><tr><td>Amount</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td><td>80 – 90</td></tr><tr><td>Frequency</td><td>5</td><td>x</td><td>15</td><td>12</td><td>7</td></tr></table>						Amount	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90	Frequency	5	x	15	12	7		
Amount	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90																
Frequency	5	x	15	12	7																
		Answer: 8																			
T	16	A streaming platform records user preference.																			
		<table><tr><td>Category Code (x)</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Frequency (f)</td><td>14</td><td>x</td><td>25</td><td>2x – 5</td><td>25</td><td>12</td></tr></table>						Category Code (x)	1	2	3	4	5	6	Frequency (f)	14	x	25	2x – 5	25	12
Category Code (x)	1	2	3	4	5	6															
Frequency (f)	14	x	25	2x – 5	25	12															
		If the distribution is bimodal, and the modal values differ by 2, find x and the modal categories.																			
		Answer: x = 14																			
H	17	An AI-based smart traffic system records the number of vehicles (in hundreds) crossing a junction during different time intervals.																			
		<table><tr><td>Vehicles (x)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Frequency (f)</td><td>8</td><td>x + 4</td><td>25</td><td>36</td><td>2x – 10</td></tr></table>						Vehicles (x)	10	20	30	40	50	Frequency (f)	8	x + 4	25	36	2x – 10		
Vehicles (x)	10	20	30	40	50																
Frequency (f)	8	x + 4	25	36	2x – 10																
		<table><tr><td></td><td>60</td><td>70</td></tr><tr><td></td><td>36</td><td>12</td></tr></table>							60	70		36	12								
	60	70																			
	36	12																			
		If the distribution is bimodal, find:																			
		1) the value of x																			
		2) both modal values																			
		3) the common modal frequency																			
		4) the total frequency																			
		5) What conclusions can be drawn regarding:																			
		• traffic congestion, • rush-hour movement, • and road management planning?																			

T	18	An AI-based online learning platform records the daily screen time (in minutes) of students attending live coding sessions. The following data was collected: <table><tr><td>Screen Time (min)</td><td>0 – 19</td><td>20 – 39</td><td>40 – 59</td><td>60 – 79</td></tr><tr><td>Frequency</td><td>6</td><td>14</td><td>28</td><td>42</td></tr><tr><td></td><td>80 – 99</td><td>100 – 119</td><td></td><td></td></tr><tr><td></td><td>38</td><td>12</td><td></td><td></td></tr></table> Answer the following: 1) Convert the inclusive class intervals into exclusive class intervals. 2) Identify the modal class. 3) Find the mode of the distribution. 4) What conclusions can be drawn about student engagement and peak learning duration from the above data?	Screen Time (min)	0 – 19	20 – 39	40 – 59	60 – 79	Frequency	6	14	28	42		80 – 99	100 – 119				38	12		
Screen Time (min)	0 – 19	20 – 39	40 – 59	60 – 79																		
Frequency	6	14	28	42																		
	80 – 99	100 – 119																				
	38	12																				
T	19	A gaming company recorded the number of hours employees spent on an AI simulation platform. <table><tr><td>Hours x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>f</td><td>5</td><td>9</td><td>x</td><td>7</td><td>3</td></tr></table> If 6 is the mode and the distribution has a unique mode, find the smallest possible value of x. Answer: x = 10	Hours x	2	4	6	8	10	f	5	9	x	7	3								
Hours x	2	4	6	8	10																	
f	5	9	x	7	3																	
T	20	An AI company records the daily cloud storage usage (in GB) of users. <table><tr><td>Storage Usage (GB)</td><td>0 – 20</td><td>20 – 40</td><td>40 – 60</td><td>60 – 80</td><td>80 – 100</td></tr><tr><td>Frequency (f)</td><td>8</td><td>15</td><td>x</td><td>28</td><td>12</td></tr></table> If the mode of the distribution is 70 GB, find the missing frequency x. Answer: x = 12	Storage Usage (GB)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	Frequency (f)	8	15	x	28	12								
Storage Usage (GB)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100																	
Frequency (f)	8	15	x	28	12																	

Unit – 1 Descriptive Statistics

T	21	A smart fitness company monitored the daily exercise time (in minutes) of employees using an AI health tracker. <table><tr><td>Exercise Time (min)</td><td>0 – 20</td><td>20 – 40</td><td>40 – 60</td><td>60 – 80</td><td>80 – 100</td></tr><tr><td>Number of Employees</td><td>12</td><td>x</td><td>30</td><td>y</td><td>30</td></tr></table> Further analysis showed that: • The distribution is bimodal. • The two modes are 50 minutes and 70 minutes. • The frequency of the second modal class is 6 more than the first modal class. Find the missing frequencies x and y. Answer: $x = y = 36$	Exercise Time (min)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	Number of Employees	12	x	30	y	30
Exercise Time (min)	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100									
Number of Employees	12	x	30	y	30									
T	22	Construct a grouped frequency distribution such that: • modal class is 40–50 • mode lies between 44 and 46 • distribution is bimodal												
C	23	If mean is 16 and median is 20. Calculate the mode. Answer: 28												
H	24	A startup company found that: Mean salary = ₹ 52,000, Median salary = ₹ 48,000 Find the mode of the salary distribution using the empirical formula. Answer: $Z = ₹ 40,000$												
T	25	For a distribution: Mean = $3x - 2$, Median = $2x + 4$ and Mode = $x + 10$ Using the empirical relation, find: x, Mean, Median and Mode. Answer: $\bar{x} = M = Z = 16$												

Method 3 $\rightsquigarrow$ Measures of Dispersion

Introduction

→ Suppose three classes have the same average marks.

Class	Marks
Class A	48, 49, 50, 51, 52
Class B	10, 30, 50, 70, 90
Class C	0, 0, 50, 100, 100

→ In all three classes, Arithmetic Mean = 50

- But are all the classes really similar?
Absolutely not!
- The **variation** in marks is completely different.

Observations

- **Class A**
 - Marks are very close to 50.
 - Students performed almost equally.
 - There is very **little** variation.
- **Class B**
 - Marks are spread far from 50.
 - Some students scored low, some very high.
 - There is **moderate** variation.
- **Class C**
 - Marks are extremely spread out.
 - Some students failed badly while some got full marks.
 - There is very high variation.

Unit – 1 Descriptive Statistics

Conclusion

- The datasets can have:
 - the same mean,
 - but completely different behavior.
- So, knowing only the average is not enough.
- We must also study:
 - how much the data spreads,
 - how far observations are from the center,
 - and how scattered the data is.
- This spread or scattering of data is called Dispersion.

What are a Measures of Dispersion?

- Measures of Dispersion, also called **Measures of Variation** are the statistical measures that show how much the data values are spread out or scattered around a central value (mean, median, or mode).
- Measures of Dispersion help us answer questions like:
 - How **stable** is the data?
 - How **consistent** are the observations?
 - How much **variability** exists in the data?
 - How **far** are the **observations** from the central value?
 - How **reliable** is the average?
- **Higher Dispersion**
 - Data values are widely scattered.
 - Observations are less consistent.
- **Lower Dispersion**
 - Data values are close to each other.
 - Observations are more consistent.

Unit – 1 Descriptive Statistics

→ Measures of dispersion are broadly classified into two categories:

- Absolute Measures
- Relative Measures

→ **Absolute Measures**

- Absolute measures express the spread or scattering of data in the same units as the original data.
- For example:
 - If the data is in cm, dispersion is also in cm.
 - If the data is in kg, dispersion is also in kg.
- Therefore, they are not suitable for comparing two datasets having different units.

→ Absolute measures we are going to study

- Range
- Mean Deviation
- Variance
- Standard Deviation

Unit – 1 Descriptive Statistics

Method 3.1 $\rightsquigarrow$ Range

→ The simplest measure of dispersion, defined as the difference between the largest and smallest values in a dataset.

$$\text{Range} = \text{Largest value} - \text{Smallest value}$$

→ Range for Individual (Ungrouped) Data

- Example:
 - 12, 18, 25, 30, 45, 50
 - Largest value = 50, Smallest Value = 12
 - Range = 50 – 12 = 38

→ Range for Discrete Frequency Distribution

Marks	Frequency
10	2
15	5
20	8
25	4
30	1

- Important Observation
 - Frequencies do not affect the range directly.
 - Only smallest and largest observations matter.
- Largest value = 30 and smallest value = 10
- Range = 30 – 10 = 20

→ Range for Continuous Frequency Distribution

Class Interval	Frequency
0 – 10	4
10 – 20	7
20 – 30	12
30 – 40	6
40 – 50	3

Unit – 1 Descriptive Statistics

→ Range is calculated by:

- subtracting the starting value of the first-class interval from the ending value of the last class interval.

→ Starting value of first-class interval = 0, Ending value of last class interval = 50

→ Range = $50 - 0 = 50$

Remark

→ Small Range → Small variation (data values are close together)

→ Large Range → High variation (data values are widely spread out)

Example of Method – 3.1: Range

C	1	A cybersecurity company monitored the response time (in milliseconds) of its AI-based firewall during different cyberattacks. The recorded response times were: 18, 22, 25, 19, 21, 24, 20, 450 (1) Find the Range of the data. (2) Is the range giving a realistic idea about the overall data? (3) What can you conclude about the effect of extreme values on Range? Answer: (1) Range = 432, (2) the range suggests that the data is highly spread out, even though most observations are actually close together. (3) Range is highly affected by extreme values (outliers).														
H	2	The daily downloads of an app are 520, 610, 730, 680, 560, 920, 780 Find the range. Answer: Range = 400														
H	3	<table border="1"><tr><td>x</td><td>12</td><td>15</td><td>18</td><td>21</td><td>24</td><td>27</td></tr><tr><td>f</td><td>4</td><td>6</td><td>10</td><td>8</td><td>5</td><td>2</td></tr></table> Find the range. Answer: 15	x	12	15	18	21	24	27	f	4	6	10	8	5	2
x	12	15	18	21	24	27										
f	4	6	10	8	5	2										

H	4	Construct a dataset of five observations having range 25.												
H	5	During an IPL season, a batsman scored the following runs in different matches: 12, 15, 18, 20, 22, 19, 105 (1) Find the range of the scores. (2) Does the range represent the overall performance properly? (3) What can you conclude about the effect of one exceptional performance on range?												
T	6	A fitness company recorded the heart rate (in beats per minute) of users during a workout session: <table><tr><td>Heart Rate (bpm)</td><td>60 – 80</td><td>80 – 100</td><td>100 – 120</td><td>120 – 140</td><td>140 – 160</td></tr><tr><td>Number of Users</td><td>2</td><td>3</td><td>0</td><td>14</td><td>8</td></tr></table> (1) Find the range of the heart rates. (2) What does the range tell us about the workout intensity? (3) In which interval are most users concentrated? (4) If one athlete recorded a heart rate of 210 bpm, how would it affect the range?	Heart Rate (bpm)	60 – 80	80 – 100	100 – 120	120 – 140	140 – 160	Number of Users	2	3	0	14	8
Heart Rate (bpm)	60 – 80	80 – 100	100 – 120	120 – 140	140 – 160									
Number of Users	2	3	0	14	8									
T	7	Can the range be zero? If yes, under what condition?												
T	8	Construct two different datasets each having range 40 but different means.												
T	9	The frequency distribution of the marks obtained by students is <table><tr><td>Marks (x)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Frequency (f)</td><td>k</td><td>8</td><td>12</td><td>5</td><td>k – 1</td></tr></table> Determine the range when $k = 0$, $k = 1$ and $k = 5$. Also comment on how the value of k affects the range.	Marks (x)	10	20	30	40	50	Frequency (f)	k	8	12	5	k – 1
Marks (x)	10	20	30	40	50									
Frequency (f)	k	8	12	5	k – 1									
T	10	Can two discrete distributions have the same range but different means? Illustrate with examples.												

T	11	A discrete distribution has a range of 40. Find the new range if: (1) Every observation is increased by 15. (2) Every observation is decreased by 15. (3) Every observation is multiplied by 5. (4) Every observation is divided by 4. (5) Every observation is multiplied by –3.
---	----	--

Method 3.2 $\rightsquigarrow$ Mean Deviation

→ Before understanding mean deviation, we must first understand deviation.

→ A deviation tells us: **How far a value is from a central value.**

- Usually, the central value is the mean.

Deviation = $x - \bar{x}$; where x is a data point

→ **Example:**

- Suppose marks of students are: 10, 12, 14, 16, 18
- $\bar{x} = 14$

Value (x)	Mean ($\bar{x} = 14$)	Deviation ($x - \bar{x}$)
10	14	-4
12	14	-2
14	14	0
16	14	+2
18	14	+4

- 10 is 4 unit below the average → deviation = -4
- 18 is 4 unit above the average → deviation = +4

→ **Important Observation**

- If we add all deviations from the mean:

$$(-4) + (-2) + 0 + 2 + 4 = 0$$

- This always happens for mean.
- Thus, **sum of all deviations from the mean is always zero.**

$$\text{i.e., } \sum (x - \bar{x}) = 0$$

- So, statisticians use **absolute values** to avoid cancellation.

Mean Deviation

→ The Mean Deviation (also called the Mean Absolute Deviation or MD) is the **average of absolute deviations** of observations from a **central value**.

→ The central value may be:

- Arithmetic Mean
- Median
- Mode

→ Table of different formulae of mean deviation:

Method	Ungrouped Data	Grouped Data
M.D. about Mean	$\frac{\sum x_i - \bar{x} }{n}$	$\frac{\sum f_i x_i - \bar{x} }{\sum f_i}$
M.D. about Median	$\frac{\sum x_i - M }{n}$	$\frac{\sum f_i x_i - M }{\sum f_i}$
M.D. about Mode	$\frac{\sum x_i - Z }{n}$	$\frac{\sum f_i x_i - Z }{\sum f_i}$

→ Remarks

- For Mean Deviation, median is always the best center.
 - Because MD uses absolute deviations: $|x - a|$ and mathematically $\sum |x - a|$ is minimum when $a = \text{median}$.
- Smaller MD $\Rightarrow$ data is more concentrated around that center.
- Larger MD $\Rightarrow$ data is more scattered from that center.

Unit – 1 Descriptive Statistics

Example of Method - 3.2: Mean Deviation

C	1	Find the mean deviation about the mean and median for the following data: 2, 4, 7, 8, 9. Answer: $MD(\bar{x}) = 2.4$, $MD(M) = 2.2$																				
H	2	Find the mean deviation about the mean, median and mode for the following data: 5, 10, 17, 20, 23, 20. Answer: $MD(\bar{x}) = 5.5556$, $MD(M) = 5.1667$, $MD(Z) = 5$																				
C	3	Find mean deviation about the mean, median and mode for the following data: <table border="1"><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>f</td><td>7</td><td>4</td><td>6</td><td>3</td><td>5</td></tr></table> Answer: $MD(\bar{x}) = 6.32$, $MD(M) = 6.2$, $MD(Z) = 9$	x	5	10	15	20	25	f	7	4	6	3	5								
x	5	10	15	20	25																	
f	7	4	6	3	5																	
H	4	Find mean deviation about the mean, median and mode for the following data: <table border="1"><tr><td>x</td><td>2</td><td>5</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>f</td><td>2</td><td>8</td><td>10</td><td>7</td><td>8</td><td>5</td></tr></table> Answer: $MD(\bar{x}) = MD(M) = MD(Z) = 2.3$	x	2	5	6	8	10	12	f	2	8	10	7	8	5						
x	2	5	6	8	10	12																
f	2	8	10	7	8	5																
T	5	A data scientist analyzed the number of hours spent by employees on an advanced cloud-computing platform during a month. <table border="1"><tr><td>Hours x</td><td>3</td><td>7</td><td>11</td><td>14</td><td>18</td></tr><tr><td>f</td><td>5</td><td>9</td><td>14</td><td>20</td><td>16</td></tr></table> <table border="1"><tr><td>21</td><td>27</td><td>35</td><td>50</td></tr><tr><td>11</td><td>7</td><td>4</td><td>2</td></tr></table> (1) Find the Mean Deviation about Mean (2) Find the Mean Deviation about Median (3) Which central value is better for this data? Answer: (1) $MD(\bar{x}) \approx 6.46$, (2) $MD(M) \approx 6.23$, (3) Median	Hours x	3	7	11	14	18	f	5	9	14	20	16	21	27	35	50	11	7	4	2
Hours x	3	7	11	14	18																	
f	5	9	14	20	16																	
21	27	35	50																			
11	7	4	2																			

C	6	A smart hospital recorded the patient waiting time (in minutes) before treatment in the emergency ward: <table><tr><td>Waiting Time (min)</td><td>5 – 25</td><td>25 – 45</td><td>45 – 65</td><td>65 – 85</td><td>85 – 105</td></tr><tr><td>Number of Patients</td><td>12</td><td>8</td><td>14</td><td>20</td><td>6</td></tr></table> Find Mean Deviation about Mean, Median and Mode. Interpret the results and determine: • Which measure of central tendency best represents the patient waiting-time data? • Whether the emergency response system is consistent or highly variable. Answer: $MD(\bar{x}) = 21.33$, $MD(M) = 21.904$, $MD(Z) = 23.466$	Waiting Time (min)	5 – 25	25 – 45	45 – 65	65 – 85	85 – 105	Number of Patients	12	8	14	20	6
Waiting Time (min)	5 – 25	25 – 45	45 – 65	65 – 85	85 – 105									
Number of Patients	12	8	14	20	6									
H	7	Find mean deviation about the mean, median and mode for the following data: <table><tr><td>Class</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>f</td><td>5</td><td>10</td><td>20</td><td>9</td><td>6</td></tr></table> Answer: $MD(\bar{x}) = 8.28$, $MD(M) = 8.20$, $MD(Z) = 8.30$	Class	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	f	5	10	20	9	6
Class	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60									
f	5	10	20	9	6									
H	8	Two companies have the same average salary: <table><tr><td>Company</td><td>Mean Salary</td><td>Mean Deviation</td></tr><tr><td>A</td><td>₹50,000</td><td>₹2000</td></tr><tr><td>B</td><td>₹50,000</td><td>₹12000</td></tr></table> Which company has more consistent salaries? Why? Answer: Company A is more consistent because it has the smaller mean deviation.	Company	Mean Salary	Mean Deviation	A	₹50,000	₹2000	B	₹50,000	₹12000			
Company	Mean Salary	Mean Deviation												
A	₹50,000	₹2000												
B	₹50,000	₹12000												

H	9	The mean deviation about median of the following distribution is 6. Find missing frequency x. <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td></tr><tr><td>f</td><td>3</td><td>x</td><td>5</td><td>2</td></tr></table> Answer: $x = 0$	x	5	10	15	20	f	3	x	5	2		
x	5	10	15	20										
f	3	x	5	2										
T	10	A data analyst calculated the Mean Deviation about Mean for the following distribution: <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>y</td><td>4</td><td>7</td><td>10</td><td>6</td><td>3</td></tr></table> Later, it was discovered that frequency of value 25 was wrongly taken as 3 instead of 8. Find: (1) Incorrect Mean Deviation about Mean (2) Correct Mean Deviation about Mean (3) Comment on the effect of the correction.	x	5	10	15	20	25	y	4	7	10	6	3
x	5	10	15	20	25									
y	4	7	10	6	3									
T	11	A music streaming company analyzed the daily listening time (in minutes) of users listening to a newly released album. <table><tr><td>Listening Time (min)</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>Number of Users</td><td>4</td><td>9</td><td>15</td><td>12</td><td>5</td></tr></table> While calculating the Mean Deviation about Mean, the analytics manager mistakenly entered the last class interval 50–60 as 80–90 due to an editing error in the entertainment dashboard. Find: (1) Incorrect Mean Deviation about Mean (2) Correct Mean Deviation about Mean (3) Explain why this mistake caused a major change in the dispersion of listening-time data.	Listening Time (min)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	Number of Users	4	9	15	12	5
Listening Time (min)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60									
Number of Users	4	9	15	12	5									

Method 3.3 $\rightsquigarrow$ Standard Deviation

Standard Deviation

→ Standard Deviation (SD) is the positive square root of the mean of squared deviations from arithmetic mean.

- It measures: **How much the data values typically spread around the mean.**

- Small SD → data is close to the mean
- Large SD → data is widely spread from the mean

→ It is denoted by “ σ ” and read as “sigma”.

→ It is defined as follow:

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

→ Table of different formulae of standard deviation

Method	Ungrouped Data	Discrete Grouped Data	Continuous Grouped Data
Direct Method	$\sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$	$\sqrt{\frac{\sum f_i x_i^2}{\sum f_i} - \left(\frac{\sum f_i x_i}{\sum f_i}\right)^2}$	
Assumed Mean Method	$\sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2}$	$\sqrt{\frac{\sum f_i d_i^2}{\sum f_i} - \left(\frac{\sum f_i d_i}{\sum f_i}\right)^2}$	
Step Deviation Method	-----	-----	$\sqrt{\frac{\sum f_i u_i^2}{\sum f_i} - \left(\frac{\sum f_i u_i}{\sum f_i}\right)^2} \times c$

→ n = total number of observations

→ In case of **continuous frequency distribution**,

x_i = mid value of the respective class.

→ In case of **assumed mean method**, A can be any value from x_i .

→ Use below formula to calculate d_i & u_i

$$d_i = x_i - A ; \quad u_i = \frac{x_i - A}{c}$$

Example of Method - 3.3: Standard Deviation

C	1	Find the standard deviation for the following data: 6, 7, 10, 12, 13, 4, 8, 12. Answer: 3.0414
H	2	Two investments give average yearly profit of ₹10,000. - Investment A: 9500, 10000, 10500 - Investment B: 1000, 19000, 10000 Find the SD and determine which investment is riskier. Answer: Investment A $\rightarrow$ 408.25, Investment B $\rightarrow$ 7,348.47
T	3	Find the Standard Deviation of: 20, 21, 22, 23, 100. What effect does the extreme value have on SD? Answer: 31.42
T	4	Without complete calculation, identify which dataset has larger SD. Dataset A: 49, 50, 51 Dataset B: 10, 50, 90 Give reason logically.
T	5	The Standard Deviation of a dataset is 5. If 10 is added to every observation, what will be the new SD? Explain logically.
T	6	The mean of five observations is 20 and the standard deviation is $\sqrt{8}$. Four observations are 16, 18, 20, and 24. Find the fifth observation. Answer: 22
T	7	Find the Standard Deviation of: -5, -3, -1, 1, 3. Show that SD can still be positive.
T	8	The mean and SD of 50 observations were found to be 40 and 5 respectively. Later it was discovered that one observation 60 was wrongly written as 80. Find the corrected: Mean and Standard Deviation. Answer: $\bar{x} = 39.6$, $\sigma = 0.92$

C	9	Find the standard deviation for the following distribution:												
<table><tr><td>x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>f</td><td>3</td><td>4</td><td>2</td><td>1</td><td>6</td></tr></table>			x	2	4	6	8	10	f	3	4	2	1	6
x	2	4	6	8	10									
f	3	4	2	1	6									
Answer: 3.1795														
H	10	Find the standard deviation for the following distribution:												
<table><tr><td>x</td><td>5</td><td>15</td><td>25</td><td>30</td></tr><tr><td>f</td><td>2</td><td>1</td><td>1</td><td>3</td></tr></table>			x	5	15	25	30	f	2	1	1	3		
x	5	15	25	30										
f	2	1	1	3										
Answer: 10.6104														
H	11	Two investments recorded yearly returns.												
<table><tr><td colspan="4">Investment A</td></tr><tr><td>Return (%)</td><td>8 – 10</td><td>10 – 12</td><td>12 – 14</td></tr><tr><td>Investors</td><td>5</td><td>15</td><td>5</td></tr></table>			Investment A				Return (%)	8 – 10	10 – 12	12 – 14	Investors	5	15	5
Investment A														
Return (%)	8 – 10	10 – 12	12 – 14											
Investors	5	15	5											
<table><tr><td colspan="4">Investment B</td></tr><tr><td>Return (%)</td><td>–10 – 0</td><td>0 – 20</td><td>20 – 30</td></tr><tr><td>Investors</td><td>5</td><td>15</td><td>5</td></tr></table>			Investment B				Return (%)	–10 – 0	0 – 20	20 – 30	Investors	5	15	5
Investment B														
Return (%)	–10 – 0	0 – 20	20 – 30											
Investors	5	15	5											
Find SD and identify the riskier investment.														
Answer: $\sigma_A = 1.26\%$, $\sigma_B = 9.49\%$, Investmetn B is Riskier														
H	12	A coaching institute recorded marks of students in a mathematics test.												
<table><tr><td>Marks</td><td>0 – 9</td><td>10 – 19</td><td>20 – 29</td><td>30 – 39</td><td>40 – 49</td></tr><tr><td>No. of Students</td><td>4</td><td>6</td><td>10</td><td>6</td><td>4</td></tr></table>			Marks	0 – 9	10 – 19	20 – 29	30 – 39	40 – 49	No. of Students	4	6	10	6	4
Marks	0 – 9	10 – 19	20 – 29	30 – 39	40 – 49									
No. of Students	4	6	10	6	4									
Find the Standard Deviation using assumed mean method.														
Answer: 12.11 marks														

Unit – 1 Descriptive Statistics

H	13	A packaging company recorded the weight of parcels delivered in a day. <table><tr><td>Weight (kg)</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td></tr><tr><td>No. of Parcels</td><td>4</td><td>8</td><td>12</td><td>8</td><td>4</td></tr></table> Find the Standard Deviation of parcel weights using step-deviation method in kilograms. Answer: 11.55 kg	Weight (kg)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	No. of Parcels	4	8	12	8	4
Weight (kg)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60									
No. of Parcels	4	8	12	8	4									
H	14	A hospital recorded patient waiting times. <table><tr><td>Waiting Time (min)</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td></tr><tr><td>Patients</td><td>4</td><td>8</td><td>10</td><td>6</td><td>2</td></tr></table> Find Standard Deviation using assumed mean method and comment on waiting time stability. Answer: 5.54 min.	Waiting Time (min)	15	20	25	30	35	Patients	4	8	10	6	2
Waiting Time (min)	15	20	25	30	35									
Patients	4	8	10	6	2									
H	15	A basketball player scored points in matches. Find SD using step-deviation method. <table><tr><td>Points</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Matches</td><td>2</td><td>5</td><td>8</td><td>5</td><td>2</td></tr></table> Answer: 10.87 points	Points	10	20	30	40	50	Matches	2	5	8	5	2
Points	10	20	30	40	50									
Matches	2	5	8	5	2									
T	16	The following distribution represents marks of students: <table><tr><td>Marks</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Frequency</td><td>4</td><td>6</td><td>10</td><td>8</td><td>2</td></tr></table> Later it was found that the frequency of 50 was wrongly recorded as 2 instead of 5. Find the corrected Standard Deviation. Answer: 12.28 marks	Marks	10	20	30	40	50	Frequency	4	6	10	8	2
Marks	10	20	30	40	50									
Frequency	4	6	10	8	2									

T	17	A survey was conducted on the daily internet usage (in hours) of 50 students. The following statistics were calculated from the recorded data: Number of students, $N = 50$, Mean internet usage = 6.2 hours and Standard deviation = 1.8 hours Later, it was discovered that two observations were entered incorrectly: • 14 hours was entered instead of 10 hours • 12 hours was entered instead of 8 hours Find the corrected standard deviation. Answer: 1.296 hours
---	----	---

Unit – 1 Descriptive Statistics

Method 3.4 $\rightsquigarrow$ Variance

- Variance is a measure of dispersion that tells us how far the observations are spread from the mean.
- In simple words:
 - Variance measures the average squared deviation of data values from their arithmetic mean.
- If the data values are very close to the mean → variance is small.
- If the data values are widely scattered → variance is large.
- It is denoted by capital letter “V” and defined as $V = \sigma^2$.

Variance Uses Square Units

- Since deviations are squared, the unit of variance also becomes squared.
- Example:
 - Suppose the heights of students are measured in cm.
 - If Variance = 64 then the unit becomes: 64 cm^2
 - But height is normally measured in cm, not in cm^2 .
 - So, variance becomes difficult to interpret practically.
 - Because of this, people mostly use Standard Deviation.
- To remove square units, we take the square root of variance.

$$SD = \sqrt{\text{Variance}} = \sqrt{64 \text{ cm}^2} = 8 \text{ cm}$$

which is easy to understand because it is in the original unit.

Why do mathematicians or statisticians still use variance?

- Although standard deviation is easier to understand, mathematicians and statisticians prefer variance in many calculations because:
 - Variance is mathematically more convenient because squared terms work smoothly in algebra, calculus, probability, and statistical formulas.
 - Many advanced statistical methods and machine learning models are based directly on variance.
 - Therefore, variance is mainly preferred for mathematical calculations, while standard deviation is preferred for practical interpretation.

Example of Method - 3.4: Variance

C	1	Find the standard deviation and variance for the following distribution: <table><tr><td>x</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td></tr><tr><td>f</td><td>6</td><td>14</td><td>10</td><td>8</td><td>1</td><td>3</td><td>8</td></tr></table> Answer: $\sigma = 19.6214$, $V = 385$	x	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	f	6	14	10	8	1	3	8
x	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70											
f	6	14	10	8	1	3	8											
H	2	Find the standard deviation and variance of the mark distribution of 30 students at mathematics examination in a class as below: <table><tr><td>Class</td><td>10 – 25</td><td>25 – 40</td><td>40 – 55</td><td>55 – 70</td><td>70 – 85</td><td>85 – 100</td></tr><tr><td>f</td><td>2</td><td>3</td><td>0</td><td>14</td><td>8</td><td>3</td></tr></table> Answer: $\sigma = 19.3391$, $V = 374.0008$	Class	10 – 25	25 – 40	40 – 55	55 – 70	70 – 85	85 – 100	f	2	3	0	14	8	3		
Class	10 – 25	25 – 40	40 – 55	55 – 70	70 – 85	85 – 100												
f	2	3	0	14	8	3												
T	3	<table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>Frequency</td><td>2</td><td>5</td><td>x</td><td>5</td><td>2</td></tr></table> If variance is 100, find missing frequency. Answer: $x = 12$	Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	Frequency	2	5	x	5	2				
Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50													
Frequency	2	5	x	5	2													
T	4	<table><tr><td>Income</td><td>Below 10</td><td>10 – 20</td><td>20 – 30</td><td>Above 30</td></tr><tr><td>Families</td><td>5</td><td>12</td><td>9</td><td>4</td></tr></table> Discuss whether variance can be calculated accurately.	Income	Below 10	10 – 20	20 – 30	Above 30	Families	5	12	9	4						
Income	Below 10	10 – 20	20 – 30	Above 30														
Families	5	12	9	4														
H	5	Two students have same average marks. Student A: 48, 49, 50, 51, 52 Student B: 20, 35, 50, 65, 80 Find variance and identify who is more consistent. Answer: Variance of Student A = 2, Variance of Student B = 450 Student A is more consistent.																
T	6	The mean and variance of five observations were calculated as 20 and 16 respectively. Later it was found that 18 was wrongly entered instead of 8. Find the corrected variance. Answer: $V = 40$																

Unit – 1 Descriptive Statistics

T	7	The variance of the observations 3, 5, x, 9, 11 is 8. Find x. Answer: $x = 7$
T	8	Can variance ever be negative? Explain.
T	9	When does variance become zero? Give an example.
T	10	A company wants the most consistent employee performance. Which employee should be selected based on variance?

Method 3.5 $\rightsquigarrow$ Coefficient of Variation

Why relative measures are needed?

- Measures like:
 - Range
 - Mean Deviation
 - Variance
 - Standard Deviation
- are called absolute measures of dispersion because they show the actual amount of variation in data.
- But sometimes, absolute variation alone is not enough for proper comparison.

Why?

Problem: 1 [Different Means]

→ Suppose

Dataset	Mean	SD
A	10	2
B	100	2

- Both datasets have the same standard deviation.
- But are they equally consistent? – **No**
 - In Dataset A, variation of 2 around mean 10 is large.
 - In Dataset B, variation of 2 around mean 100 is very small.
- So, standard deviation alone does not give the complete picture.
- This creates the need for a relative measure.

Unit – 1 Descriptive Statistics

Problem: 2 [Different Units]

→ Now consider another situation

Dataset	Mean	SD
Heights	170 cm	10 cm
Weights	70 kg	8 kg

→ Can we directly say:

- 10 cm SD is greater than 8 kg SD, so heights are more variable? - **No**.
- Because:
 - one dataset is measured in cm,
 - another is measured in kg.

→ Their units are completely different. Hence, direct comparison using SD becomes mathematically meaningless.

→ Therefore, we need a measure that:

- removes units,
- compares variation fairly,
- and expresses variability relative to the average.

→ This leads to the concept of a **Relative Measure** of Dispersion.

Relative Measures

→ A relative measure of dispersion is a **unit-free** measure that expresses variability in relation to a central value (usually the **mean**), showing how large the variation is compared to the average value of the data.

→ It helps in comparing datasets even when their units or magnitudes are different.

→ We use relative measures to compare consistency between different datasets.

→ Among all relative measures, the most important and widely used measure is the **Coefficient of Variation (CV)**, because it compares standard deviation directly with the mean and expresses variability in percentage form.

Coefficient of Variation

→ Coefficient of Variation is the ratio of standard deviation to arithmetic mean, expressed as a percentage.

→ Coefficient of Variance is defined as

$$C. V. = \frac{\sigma}{\bar{x}} \times 100$$

where

σ = Standard Deviation and $\bar{x}$ = Mean

→ Smaller CV → Data is more consistent, stable, and reliable.

→ Larger CV → Data is less consistent and more variable.

Explanation of the Definition

→ Ratio of Standard Deviation to Mean

→ In CV, we divide:

- Standard Deviation (SD) → which shows variation

by

- Mean → which shows the average size of the data.

→ So, CV compares:

- How large the variation is compared to the average value.

→ This is why CV is called a relative measure.

Most Important Feature of CV

→ When we divide:

$$\frac{\text{Mean}}{\text{SD}}$$

the units cancel out.

→ **Example:**

$$\frac{170 \text{ cm}}{10 \text{ cm}}$$

cm cancels.

Unit – 1 Descriptive Statistics

→ Similarly,

$$\frac{8 \text{ kg}}{70 \text{ kg}}$$

kg cancels.

→ So, CV becomes: unit free and pure percentage.

→ That is why datasets with different units can now be compared fairly.

Example of Method - 3.5: Coefficient of Variation

C	1	The arithmetic means of runs scored by three batsmen A,B and C, in the same series of 10 innings, are 50,48 and 12 respectively. The standard deviations of their runs are 15,12 and 2 respectively. Who is the most consistent of the three? Answer: Batsman C is most consistent.																				
H	2	Runs scored by two batsmen A, B in 9 consecutive matches is given below: <table><tr><td>A</td><td>85</td><td>20</td><td>62</td><td>28</td><td>74</td><td>5</td><td>69</td><td>4</td><td>13</td></tr><tr><td>B</td><td>72</td><td>4</td><td>15</td><td>30</td><td>59</td><td>15</td><td>49</td><td>27</td><td>26</td></tr></table> Which of the batsman is more consistent? Answer: Batsman B is more consistent.	A	85	20	62	28	74	5	69	4	13	B	72	4	15	30	59	15	49	27	26
A	85	20	62	28	74	5	69	4	13													
B	72	4	15	30	59	15	49	27	26													
T	3	Goals scored by two team A and B in a football season were as shown in the table. Find out which team is more consistent. <table><tr><td>Number of goals in a match</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Team A</td><td>27</td><td>9</td><td>8</td><td>5</td><td>4</td></tr><tr><td>Team B</td><td>17</td><td>9</td><td>6</td><td>5</td><td>3</td></tr></table> Answer: Team B is more consistent.	Number of goals in a match	0	1	2	3	4	Team A	27	9	8	5	4	Team B	17	9	6	5	3		
Number of goals in a match	0	1	2	3	4																	
Team A	27	9	8	5	4																	
Team B	17	9	6	5	3																	

C	4	Two machines A, B are used to fill a mixture of cement concrete in a beam. Find the standard deviation of each machine & comment on the performances of two machines. <table><tr><td>A</td><td>32</td><td>28</td><td>47</td><td>63</td><td>71</td><td>39</td><td>10</td><td>60</td><td>96</td><td>14</td></tr><tr><td>B</td><td>19</td><td>31</td><td>48</td><td>53</td><td>67</td><td>90</td><td>10</td><td>62</td><td>40</td><td>80</td></tr></table> Answer: $\sigma_A = 25.4950$, $\sigma_B = 24.4294$ There is less variability in the performance of the machine B	A	32	28	47	63	71	39	10	60	96	14	B	19	31	48	53	67	90	10	62	40	80
A	32	28	47	63	71	39	10	60	96	14														
B	19	31	48	53	67	90	10	62	40	80														
C	5	An analysis of monthly wages paid to the workers of two firms A and B belonging to the same industry gives the following results: <table><tr><td></td><td>Firm A</td><td>Firm B</td></tr><tr><td>Number of workers</td><td>500</td><td>600</td></tr><tr><td>Average daily wage</td><td>186</td><td>175</td></tr><tr><td>Variance of distribution of wages</td><td>81</td><td>100</td></tr></table> (1) Which firm has a larger wage bill? (2) In which firm, is there greater variability in individual wages? (3) Calculate average daily wages of all the workers in the firms A & B taken together. Answer: (1) Firm B, (2) Firm B, (3) 180		Firm A	Firm B	Number of workers	500	600	Average daily wage	186	175	Variance of distribution of wages	81	100										
	Firm A	Firm B																						
Number of workers	500	600																						
Average daily wage	186	175																						
Variance of distribution of wages	81	100																						
H	6	<table><tr><td>Company</td><td>Average Salary</td><td>SD</td></tr><tr><td>X</td><td>₹40,000</td><td>₹8,000</td></tr><tr><td>Y</td><td>₹80,000</td><td>₹12,000</td></tr></table> Which company has more stable salary distribution? Answer: Company Y	Company	Average Salary	SD	X	₹40,000	₹8,000	Y	₹80,000	₹12,000													
Company	Average Salary	SD																						
X	₹40,000	₹8,000																						
Y	₹80,000	₹12,000																						
H	7	A dataset has: Mean = 80 and CV = 25% Find the standard deviation. Answer: 20																						

Unit – 1 Descriptive Statistics

H	8	A cloud server recorded daily response times (in seconds): 1, 25, 2, 30, 1, 35, 2 Find CV. Why is CV extremely large? Comment on reliability of the server. Answer: CV = 104.7%												
T	9	A researcher calculated: Mean = 40, Variance = 64 and CV = 20% Later, it was found that one observation “70” was wrongly entered instead of “30”. If total observations = 20, find corrected mean, corrected variance and corrected CV. Answer: $\bar{x} = 38$, $V = 20$, $CV = 11.77\%$												
T	10	Which affects CV more? changing mean slightly, or changing SD slightly? Explain conceptually.												
T	11	Explain why CV is called a unit-free measure.												
T	12	A dataset has Mean = 0. Can CV be calculated? Why or why not? What practical issue occurs here?												
T	13	A company compares stability of three delivery systems. <table border="1"> <thead> <tr> <th>System</th><th>Mean Delivery Time</th><th>SD</th></tr> </thead> <tbody> <tr> <td>Drone</td><td>15 Min</td><td>6</td></tr> <tr> <td>Bike</td><td>40 Min</td><td>8</td></tr> <tr> <td>Truck</td><td>100 Min</td><td>12</td></tr> </tbody> </table> (1) Which system is most reliable? (2) Which system fluctuates most in absolute terms? (3) Which system fluctuates most relative to its average? (4) Why are SD and CV giving different impressions?	System	Mean Delivery Time	SD	Drone	15 Min	6	Bike	40 Min	8	Truck	100 Min	12
System	Mean Delivery Time	SD												
Drone	15 Min	6												
Bike	40 Min	8												
Truck	100 Min	12												
T	14	A researcher claims: “If all observations of a dataset are multiplied by 10, then its CV also becomes 10 times.” Is the statement true or false? Justify mathematically without using actual data.												

- T 15** Lives of two models of refrigerators turned in for new models in a recent survey are given in the adjoining table.

Life (in year)	0 – 2	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12
Model A	5	16	13	7	5	4
Model B	2	7	12	19	9	1

- (1) What is the average life of each model of these refrigerators?
 (2) Which model shows more uniformity?

Answer: (1) 5.12 & 6.16, (2) Model B

- T 16** A fitness training academy conducted a physical endurance test for students. The time taken (in minutes) to complete an obstacle course was recorded.

Completion Time (min)	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of Students	2	x	6	y	2

The academy reported that:

- Total number of students tested = 20
- Average completion time = 25 minutes

The academy now wants to analyze the consistency of students' performance for selecting candidates for an advanced training program.

- (1) Find the missing frequencies x and y.
 (2) Calculate the Standard Deviation using step-deviation method.
 (3) Based on the standard deviation, comment whether:
- students' performance is highly consistent,
 - moderately consistent, or
 - highly variable.
- (4) Which type of students would the academy prefer for the advanced program:
- students with low variation in completion time, or
 - students with high variation? Give reason.

Answer: (1) $x = y = 5$, (2) $\sigma = 11.40$ Minutes (3) $CV = 45.61\%$

Method 4 $\rightsquigarrow$ Moments

Moments

→ A moment is a numerical measure that describes a dataset by considering the deviations of its observations from a chosen reference point.

→ Moments help us understand:

- Where the data is located
- How much the data varies
- Whether the data is balanced or tilted to one side
- Whether extreme values (outliers) are likely to occur

General Formula of a Moment

→ A moment is the arithmetic mean of the powers of the deviations of observations from a chosen reference point.

$$\text{Moment} = \frac{\sum (x_i - a)^r}{n}$$

→ where

- x = observation
- a = reference point
- r = order of moment

Moment about Mean

→ If deviations of data are taken from **mean**, then it is known as moment about mean.

- Reference point: $a = \bar{x}$

→ It is also known as **central moment**.

→ It is denoted by " μ_r ".

→ First Moment about actual mean is always zero.

→ Second Moment about actual mean is variance.

Moment about Assumed Mean

→ If deviations of data are taken about any assumed value(a) from x_i , then it is known as moment about assumed mean.

→ It is denoted by " μ'_r ".

Unit – 1 Descriptive Statistics

→ These moments are mainly used to simplify calculations.

Moment about Zero

→ If deviations of data are taken about zero, it is known as moment about zero.

- Reference point: **a = 0**
- These are called Raw Moments or Moments About Origin.

→ It is denoted by " **v_r** ".

→ The first moment about zero is actual mean.

→ Table of different formulae of moment.

Moment about	Ungrouped Data	Grouped Data
Mean (μ_r)	$\frac{\sum (x_i - \bar{x})^r}{n}$	$\frac{\sum f_i (x_i - \bar{x})^r}{\sum f_i}$
Assumed mean (μ'_r)	$\frac{\sum (x_i - a)^r}{n}$	$\frac{\sum f_i (x_i - a)^r}{\sum f_i}$
Zero (v_r)	$\frac{\sum x_i^r}{n}$	$\frac{\sum f_i x_i^r}{\sum f_i}$

where, $r = 1, 2, 3, 4, \dots$

Example of Method - 4: Moments

C	1	Find the first four moments about assumed mean 5, actual mean and zero for the data 1, 3, 7, 9, 10.
		Answer: $\mu = 0, \quad 12, \quad -12, \quad 208.8$ $\mu' = 1, \quad 13, \quad 25, \quad 233.8$ $v = 6, \quad 48, \quad 420, \quad 3808.8$

Unit – 1 Descriptive Statistics

H	2	Find the first four moments about assumed mean 14, actual mean and zero for the data 11, 12, 14, 16, 20. Answer: $\mu = 0, \quad 10.24, \quad 19.162, \quad 213.5872$ $\mu' = 0.6, \quad 10.6, \quad 37.8, \quad 281.8$ $v = 14.6, \quad 223.4, \quad 3579.8, \quad 57225.8$
H	3	The daily screen times (hours) of six students are: 2, 3, 4, 5, 6, 12 Find Mean, second central moment and third central moment. Answer: $\bar{x} = 5.333$ hours, $\mu_2 = 10.556$ hours², $\mu_3 = 40.741$ hours³
T	4	An e-commerce company records delivery times (days) for 8 orders: 2, 3, 4, 3, 5, 4, 6, 3. Find first and second moment about 3. Interpret the meaning of the second moment. Answer: $\mu'_1(3) = 0.75, \quad \mu'_2(3) = 2$
T	4	Daily app usage (hours) of five users: 1, 2, 2, 3, 7 Find: (1) First moment about zero. (2) Second moment about zero. (3) Identify whether a few heavy users dominate platform engagement. Answer: $v_1 = 3$ hours, $v_2 = 13.4$ hours²
T	6	A student's daily coding hours over 7 days are: 2, 3, 4, x, 5, 6, 8. If the first moment about the origin is 5 hours, find the missing value x. Answer: $x = 7$ hours
T	7	The daily sales (units) are: 10, 12, 13, 15, x. If the third central moment equals 180, find x. Answer: $x \approx 25.27$ units

C	8	Calculate the four moments about assumed mean 5, actual mean and zero for following distribution. <table><tr><td>x</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f</td><td>1</td><td>3</td><td>7</td><td>3</td><td>1</td></tr></table> Answer: $\mu = 0,$ $0.9333,$ $0,$ 2.5333 $\mu' = -1,$ $1.9333,$ $-3.8,$ 9.1333 $v = 4,$ $16.9333,$ $75.2,$ 348.1333	x	2	3	4	5	6	f	1	3	7	3	1
x	2	3	4	5	6									
f	1	3	7	3	1									
H	9	Calculate the four moments about assumed mean 15, actual mean and zero for following distribution. <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>f</td><td>6</td><td>10</td><td>14</td><td>6</td><td>4</td></tr></table> Answer: $\mu = 0,$ $34,$ $40.5,$ 2707 $\mu' = -1,$ $35,$ $-62.5,$ 2750 $v = 14,$ $230,$ $4212.5,$ 83375	x	5	10	15	20	25	f	6	10	14	6	4
x	5	10	15	20	25									
f	6	10	14	6	4									
H	10	A cloud service provider is evaluating the performance of one of its servers. During a monitoring period, the network latency experienced by incoming requests was recorded as follows: <table><tr><td>Latency (ms)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>No. of Requests</td><td>5</td><td>12</td><td>18</td><td>10</td><td>5</td></tr></table> Calculate the first four central moments and state the units of each moment. Answer: $\mu_1 = 0 \text{ ms},$ $\mu_2 = 123.84 \text{ ms}^2,$ $\mu_3 = 108.672 \text{ ms}^3,$ $\mu_4 = 36400 \text{ ms}^4$	Latency (ms)	10	20	30	40	50	No. of Requests	5	12	18	10	5
Latency (ms)	10	20	30	40	50									
No. of Requests	5	12	18	10	5									

Unit – 1 Descriptive Statistics

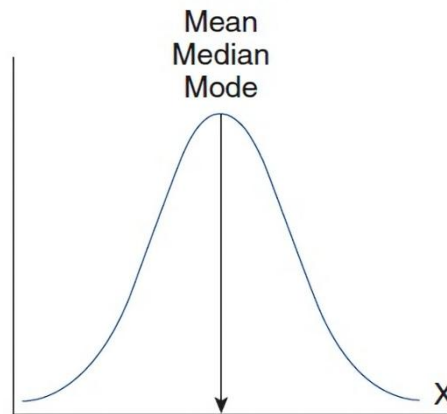
T	11	A smartphone company records the number of battery replacements required during the warranty period. <table><tr><td>Replacements</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Number of Phones</td><td>20</td><td>15</td><td>k</td><td>5</td></tr></table> The company found that the second moment about the origin exceeds the first moment about the origin by 1 unit. Determine the missing frequency k. Answer: k = 10	Replacements	0	1	2	3	Number of Phones	20	15	k	5		
Replacements	0	1	2	3										
Number of Phones	20	15	k	5										
T	12	A cloud server recorded latency values (ms). <table><tr><td>Latency (ms)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>Frequency</td><td>5</td><td>10</td><td>k</td><td>10</td><td>5</td></tr></table> If the fourth central moment is 45,000 ms⁴, find k. Answer: k = 10	Latency (ms)	10	20	30	40	50	Frequency	5	10	k	10	5
Latency (ms)	10	20	30	40	50									
Frequency	5	10	k	10	5									
C	13	Calculate the moments about assumed mean 25, actual mean and zero for following distribution: <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td></tr><tr><td>f</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table> Answer: $\mu = 0$, 81, –144, 14817 $\mu' = -3$, 90, –900, 21000 $v = 22$, 565, 15850, 471625	Class	0 – 10	10 – 20	20 – 30	30 – 40	f	1	3	4	2		
Class	0 – 10	10 – 20	20 – 30	30 – 40										
f	1	3	4	2										
H	14	Calculate the moments about assumed mean 65, actual mean and zero for following distribution: <table><tr><td>Class</td><td>60 – 62</td><td>63 – 65</td><td>66 – 68</td><td>69 – 71</td><td>72 – 74</td></tr><tr><td>f</td><td>5</td><td>18</td><td>42</td><td>27</td><td>8</td></tr></table> Answer: $\mu = 0$, 8.5275, –2.6933, 199.3759 $\mu' = 2.45$, 14.53, 74.69, 516.13 $v = 67.45$, 4558.03, 308586.79, 20930221.03	Class	60 – 62	63 – 65	66 – 68	69 – 71	72 – 74	f	5	18	42	27	8
Class	60 – 62	63 – 65	66 – 68	69 – 71	72 – 74									
f	5	18	42	27	8									

H	15	<table><tr><td>Units Consumed</td><td>100 – 150</td><td>150 – 200</td><td>200 – 250</td></tr><tr><td>f</td><td>5</td><td>10</td><td>18</td></tr><tr><td></td><td>250 – 300</td><td>300 – 350</td><td></td></tr><tr><td></td><td>12</td><td>5</td><td></td></tr></table> Taking A = 225, calculate the first three moments about the assumed mean. Answer: $\mu'_1 = 2$ unit, $\mu'_2 = 3100$ unit², $\mu'_3 = 5000$ unit³	Units Consumed	100 – 150	150 – 200	200 – 250	f	5	10	18		250 – 300	300 – 350			12	5	
Units Consumed	100 – 150	150 – 200	200 – 250															
f	5	10	18															
	250 – 300	300 – 350																
	12	5																
T	16	A startup ecosystem has the following annual revenue distribution (in ₹ lakhs): <table><tr><td>Revenue (₹ Lakhs)</td><td>0 – 20</td><td>10 – 20</td><td>20 – 30</td></tr><tr><td>Frequency</td><td>5</td><td>10</td><td>k</td></tr></table> Suppose one additional startup with annual revenue in the class 500–520 lakhs are included in the dataset. Without performing detailed calculations, discuss which of the first four central moments will be affected the most and justify your answer.	Revenue (₹ Lakhs)	0 – 20	10 – 20	20 – 30	Frequency	5	10	k								
Revenue (₹ Lakhs)	0 – 20	10 – 20	20 – 30															
Frequency	5	10	k															
T	17	Two cloud servers have the same average response time. Server A: Variance = 16 ms² and Server B: Variance = 36 ms² Using moments, determine which server provides more consistent performance. Answer: Server A provides more consistent performance.																

Method 5 $\rightsquigarrow$ Measures of Skewness

Symmetrical Distribution

→ A distribution where the **right half is a mirror image of the left half** is called symmetrical.



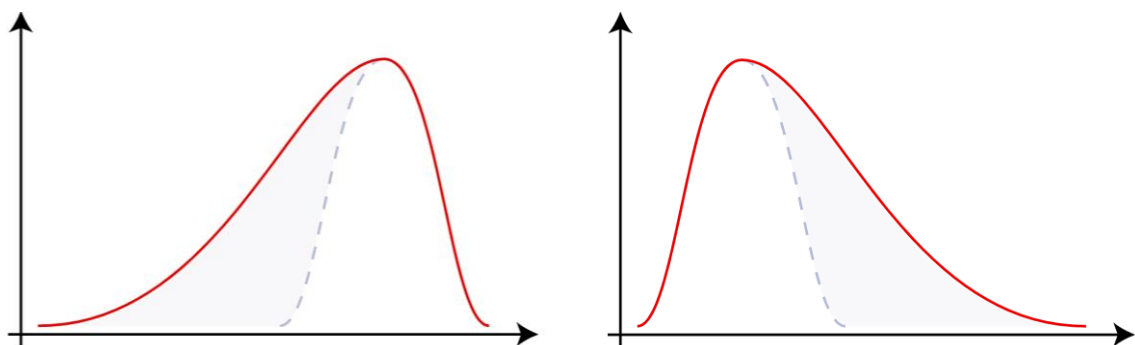
→ For symmetrical data:

$$\text{Mean} = \text{Median} = \text{Mode}$$

→ A symmetrical distribution is important because it gives a true central tendency (Mean = Median = Mode), produces balanced deviations on both sides, allows standard deviation and other statistical methods to work correctly, and prevents distortion from extreme values.

Asymmetrical Distribution

→ A distribution is said to be **asymmetrical** or **skewed** if its **mean, median, and mode is not equal**.



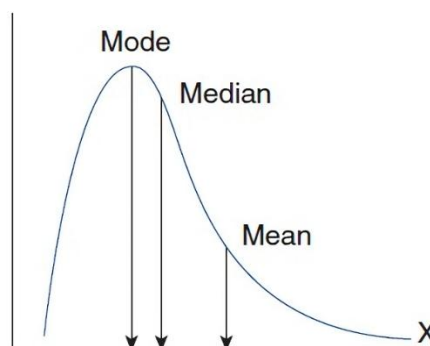
Skewness

- Skewness means **lack** of symmetry.
- Skewness is a statistical measure that describes the **asymmetry** of a data distribution.
- Skewness tells the shape of data, which mean, median, SD cannot tell.
- Shape simply means: How the data LOOKS when you draw it on a graph.
- Shape includes things like:
 - Is it balanced?
 - Is it pulled to one side?
 - Is it flat or peaked?
 - Does it have a long tail?
 - Are there extreme values?

Types of Skewness

(1) Positive Skewness

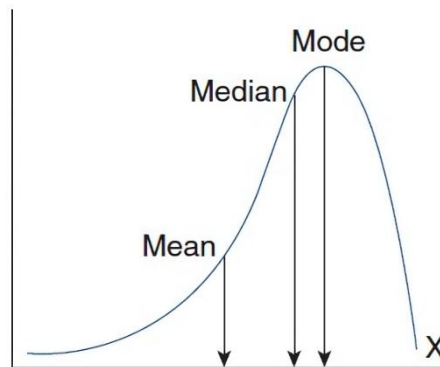
- The **right** tail is **longer**; the mass of the distribution is concentrated on the left.
 - i.e., $\mu_3 > 0 \Rightarrow \sum (x - \bar{x})^3$ is positive.
 - Mean > Median > Mode



(2) Negative Skewness

- The left tail is longer; the mass of the distribution is concentrated on the right.
 - i.e., $\mu_3 < 0 \Rightarrow \sum (x - \bar{x})^3$ is negative.
 - Mean < Median < Mode

Unit – 1 Descriptive Statistics



Measures of Skewness

→ Measure of skewness give us an idea about the extent of lopsidedness in a data.

→ The various measures of skewness are

- (1) Karl Pearson's method
- (2) Method of moments

Karl Pearson's Method

→ Karl Pearson's **coefficient of skewness** is defined as

$$\text{Skewness} = S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}} = \frac{\bar{x} - Z}{\sigma}$$

Method of Moments

→ **Measure of skewness** is obtained from moment about mean and defined as

$$\text{Skewness} = \beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3}$$

→ Remarks

- The skewness value can be either positive or negative or zero or even undefined.
- If skewness value is zero, then the distribution is known as symmetric.
- If standard deviation is zero, then the skewness is not defined.

Example of Method - 5: Measures of Skewness

C	1	Compute the Karl Pearson's coefficient of skewness for data: 25, 15, 23, 40, 27, 25, 23, 25, 20 Answer: -0.03												
H	2	The weights (kg) of employees are: 55, 58, 60, 62, 62, 62, 65, 70 Calculate Karl Pearson's coefficient of skewness. Answer: -0.059												
H	3	Find skewness by the method of moments for data: 38.2, 40.9, 39.5, 44, 39.6, 40.5, 39.5. Answer: 1.3035												
H	4	<table border="1"><tr><td>No. of Defects</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f</td><td>5</td><td>12</td><td>20</td><td>8</td><td>5</td></tr></table> Find Karl Pearson's coefficient of skewness. Answer: -0.0732	No. of Defects	0	1	2	3	4	f	5	12	20	8	5
No. of Defects	0	1	2	3	4									
f	5	12	20	8	5									
C	5	Karl Pearson's coefficient of skewness of a distribution is 0.32, its standard deviation is 6.5 and mean is 29.6. Find the mode for the distribution. Answer: 27.52												
H	6	If Mean < Median < Mode. Interpret the shape of distribution.												
T	7	Karl Pearson's coefficient of skewness of a distribution is 0.3, its variance is 8 and mean is 200. Find the mode and median for the distribution. Answer: $Z = 199.15$, $M = 199.72$												
T	8	Mean = 40, Median = 35, Mode = 30 and SD = 5. Without calculation, state: Direction of skewness. Whether distribution is positively or negatively skewed												

Unit – 1 Descriptive Statistics

T	9	A statistician calculated the following from a dataset of 20 observations: $\bar{x} = 50, \sigma = 10, Z = 46$. The Karl Pearson coefficient of skewness was therefore reported as $S_k = 0.4$. Later it was found that one observation 60 was wrongly entered as 80. Recalculate Karl Pearson's coefficient of skewness. Answer: $S_k = 0.391$								
T	10	A company reports: Mean salary = ₹ 58,000, Median salary = ₹ 50,000 and SD = ₹ 12,000. Find Karl Pearson's coefficient of skewness and interpret the salary structure. Answer: $S_k = 2$								
T	11	A dataset has $S_k = 0$. What can be concluded?								
C	12	For a group of 10 items, $\sum x = 452, \sum x^2 = 24270$ and mode is 43.7 then, find Karl Pearson's coefficient of skewness. Answer: 0.0776								
H	13	The marks obtained by a group of 25 students satisfy: $\sum x = 1125,$ The coefficient of variation is 25% and the mode is 6 marks less than the mean. Find Karl Pearson's coefficient of skewness. Answer: 0.533								
C	14	From the marks scored by 100 students in section A and 100 students in section B of a class, the following measures were obtained <table border="1"><tr><td>Section A</td><td>$\mu_A = 55$</td><td>$\sigma_A = 15.4$</td><td>Mode = 58.72</td></tr><tr><td>Section B</td><td>$\mu_B = 53$</td><td>$\sigma_B = 15.4$</td><td>Mode = 48.83</td></tr></table> Determine which distribution of marks is more skewed. Answer: Section B is more skewed	Section A	$\mu_A = 55$	$\sigma_A = 15.4$	Mode = 58.72	Section B	$\mu_B = 53$	$\sigma_B = 15.4$	Mode = 48.83
Section A	$\mu_A = 55$	$\sigma_A = 15.4$	Mode = 58.72							
Section B	$\mu_B = 53$	$\sigma_B = 15.4$	Mode = 48.83							

H	15	From the marks scored by 120 students in section A and 120 students in section B of a class, the following measures were obtained <table><tr><td>Section A</td><td>$\mu_A = 46.83$</td><td>$\sigma_A = 14.8$</td><td>Mode = 51.67</td></tr><tr><td>Section B</td><td>$\mu_B = 47.83$</td><td>$\sigma_B = 14.8$</td><td>Mode = 47.07</td></tr></table> Determine which distribution of marks is more skewed. Answer: Section A is more skewed	Section A	$\mu_A = 46.83$	$\sigma_A = 14.8$	Mode = 51.67	Section B	$\mu_B = 47.83$	$\sigma_B = 14.8$	Mode = 47.07								
Section A	$\mu_A = 46.83$	$\sigma_A = 14.8$	Mode = 51.67															
Section B	$\mu_B = 47.83$	$\sigma_B = 14.8$	Mode = 47.07															
C	16	Prove that the skewness of the following data is 0.0390 using method of moment. <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td></tr><tr><td>Frequency</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table>	Class	0 – 10	10 – 20	20 – 30	30 – 40	Frequency	1	3	4	2						
Class	0 – 10	10 – 20	20 – 30	30 – 40														
Frequency	1	3	4	2														
H	17	Find skewness of the following data using method of moment. <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td></tr><tr><td>f</td><td>8</td><td>15</td><td>20</td><td>32</td><td>23</td><td>17</td><td>5</td></tr></table> Answer: 0.01141	x	5	10	15	20	25	30	35	f	8	15	20	32	23	17	5
x	5	10	15	20	25	30	35											
f	8	15	20	32	23	17	5											
H	18	The first three central moments are: $\mu_1 = 0, \mu_2 = 16, \mu_3 = 32$. Find the moment coefficient of skewness. Answer: $\beta_1 = 0.25$																
T	19	Find Karl Pearson's coefficient of skewness and skewness based on the method of moments for the following data: <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>f</td><td>13</td><td>20</td><td>30</td><td>25</td><td>12</td></tr></table> Answer: – 0.1135, 0.0085	Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	f	13	20	30	25	12				
Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50													
f	13	20	30	25	12													
T	20	<table><tr><td>Distribution</td><td>μ_2</td><td>μ_3</td></tr><tr><td>A</td><td>16</td><td>32</td></tr><tr><td>B</td><td>25</td><td>125</td></tr></table> Which distribution is more skewed? Answer: Distribution B is more skewed.	Distribution	μ_2	μ_3	A	16	32	B	25	125							
Distribution	μ_2	μ_3																
A	16	32																
B	25	125																

T	21	If every observation is multiplied by 5, what happens to β_1 ?
T	22	If every observation is converted from meters to centimeters, what happens to β_1 ?

Method 6 $\rightsquigarrow$ Kurtosis

Introduction

- Measures of central tendency providing a summary of the overall data.
- Measures of dispersion estimate the spread or variability of data points around the central value.
- Skewness depicts whether the data is evenly distributed or tends to lean towards one side.
- Now the question arises: **How do we measure the tailedness of the distribution?**

Kurtosis

- Kurtosis is the statistical measure of **tailedness** — how often extremely large or small values occur.
- It measures **how heavy or light the tails** of a distribution are compared to a normal curve.
 - If the tails are heavy → more extreme events → high kurtosis
 - If tails are light → fewer extreme events → low kurtosis
- “ β_2 ” is the **coefficient of Kurtosis** defined as

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2}$$

- It measures how heavy the tails of a distribution are (and hence the tendency of extreme values).
- where
 - $\mu_2 = 2^{\text{nd}}$ central moment = variance
 - $\mu_4 = 4^{\text{th}}$ central moment = **tail/extreme sensitivity moment**

Types of Kurtosis

→ Leptokurtic

- Lepto - means "thin" or "slender."
- Meaning:
 - More extreme values than normal
 - fat or heavy tails
 - Outliers appear frequently

Unit – 1 Descriptive Statistics

- High risk
- Kurtosis Value: $\beta_2 > 3$

→ Mesokurtic

- **Meso** - means "middle"
- **Meaning**
 - Moderate peak and Moderate tails
 - Extreme values occur as expected
 - No unusual risk
 - Balanced shape
- **Kurtosis Value:** $\beta_2 = 3$

→ Platykurtic

- **Platy** - means "flat" (like a plate)
- **Meaning**
 - the tails are **thin**.
 - Outliers are very rare. The data is spread out, but it stays within a predictable range. There are no massive surprises
- **Kurtosis Value:** $\beta_2 < 3$

Summary

- Sometimes Statisticians subtract 3 from the calculation to make the Normal Distribution equal to 0. This is called "**Excess Kurtosis**."
- Excess Kurtosis = $\gamma_2 = \beta_2 - 3$
- Modern software (like Python, Excel, or R) often uses "**Excess Kurtosis**" instead of raw β_2 .

Type	Tail Weight	Extremes	β_2	γ_2	Risk Level
Leptokurtic	Heavy	Many	> 3	> 0	High
Mesokurtic	Normal	Normal	$= 3$	$= 0$	Neutral
Platykurtic	Light	Few	< 3	< 0	Low

Example of Method - 6: Kurtosis

H	1	Find the Kurtosis for the data 1, 3, 7, 9, 10. Also comment on type of distribution. Answer: Kurtosis = 1.45, Distribution is platykurtic												
H	2	Find the Kurtosis for the following data. Also comment on type of distribution. <table><tr><td>x</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>f</td><td>1</td><td>3</td><td>7</td><td>3</td><td>1</td></tr></table> Answer: Kurtosis = 2.9084, Distribution is platykurtic	x	2	3	4	5	6	f	1	3	7	3	1
x	2	3	4	5	6									
f	1	3	7	3	1									
C	3	Find out the kurtosis of the following data: <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>f</td><td>10</td><td>20</td><td>40</td><td>20</td><td>10</td></tr></table> Answer: 2.5	Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	f	10	20	40	20	10
Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50									
f	10	20	40	20	10									
H	4	Find the kurtosis for the following data. Also, Comment on type of distribution. <table><tr><td>Class</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td></tr><tr><td>f</td><td>1</td><td>3</td><td>4</td><td>2</td></tr></table> Answer: Kurtosis = 2.2583, Distribution is platykurtic	Class	0 – 10	10 – 20	20 – 30	30 – 40	f	1	3	4	2		
Class	0 – 10	10 – 20	20 – 30	30 – 40										
f	1	3	4	2										
T	5	<table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>Students</td><td>4</td><td>8</td><td>16</td><td>8</td><td>4</td></tr></table> Find β_2 . Answer: 11.67	Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	Students	4	8	16	8	4
Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50									
Students	4	8	16	8	4									

Unit – 1 Descriptive Statistics

T	6	Find the coefficient of variation, β_1 and β_2 for the following data: <table><tr><td>x</td><td>170 – 180</td><td>180 – 190</td><td>190 – 200</td><td>200 – 210</td></tr><tr><td>f</td><td>52</td><td>68</td><td>85</td><td>92</td></tr><tr><td></td><td>210 – 220</td><td>220 – 230</td><td>230 – 240</td><td>240 – 250</td></tr><tr><td></td><td>100</td><td>95</td><td>70</td><td>28</td></tr></table> Answer: C. V. = 9.4, $\beta_1 = 0.0034$, $\beta_2 = 2.0340$	x	170 – 180	180 – 190	190 – 200	200 – 210	f	52	68	85	92		210 – 220	220 – 230	230 – 240	240 – 250		100	95	70	28
x	170 – 180	180 – 190	190 – 200	200 – 210																		
f	52	68	85	92																		
	210 – 220	220 – 230	230 – 240	240 – 250																		
	100	95	70	28																		
T	7	A cloud computing company monitors the response time (in milliseconds) of its AI-based recommendation engine during peak traffic hours. After analyzing 50,000 user requests, the data science team computed the following central moments: $\mu_2 = 64, \mu_4 = 40,960$ The company wants to determine whether the system occasionally experiences unusually large delays that could affect user experience. (1) Find the coefficient of kurtosis. (2) Classify the distribution as Leptokurtic, Mesokurtic, or Platykurtic. (3) What does the result suggest about the occurrence of extremely slow server responses? (4) Why is kurtosis more useful than standard deviation alone for a cloud-service engineer monitoring system reliability?																				

T	8	An investment firm studies the daily percentage returns of a newly launched cryptocurrency fund. The risk analytics team obtains the following central moments from historical return data: $\mu_2 = 0.0004, \mu_4 = 0.00000256$ Investors are concerned about the possibility of sudden market crashes or unusually large gains. Compute the coefficient of kurtosis. Identify the nature of the return distribution. Based on the kurtosis value, comment on the likelihood of extreme profit or loss days. Two mutual funds have the same mean return and standard deviation, but Fund A has the above kurtosis while Fund B has $\beta_2 = 3$. Which fund would a risk-averse investor prefer? Justify. Explain why many financial institutions monitor kurtosis in addition to volatility when assessing market risk.
T	9	A social media company analyzes daily views of videos. Most videos receive 500–1000 views, but a few suddenly get millions. The view-count distribution has $\beta_2 = 25$. (1) Why is kurtosis extremely high? (2) What does it say about viral content? (3) Why would mean views not tell the complete story?

***** End of the Unit *****